

A Complete Proof of the Goodman–Style Lower Bound for Odd-Cycle Densities in Graphons

Consolidated working note for H. Yang and collaborators

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Abstract

Let $W : [0, 1]^2 \rightarrow [0, 1]$ be a graphon of edge density p , and let $m \geq 3$ be odd. We prove the conjectured Goodman–style inequality

$$t(C_m, W) \geq p^m - p(1 - p)^{m-1}$$

for every $p \in [0, 1]$. The proof is divided at $p = 2/3$, or equivalently at the complement density $q = 1/3$.

For $p \geq 2/3$, a two-sided Schur-complement identity cancels the unknown odd moments of the compressed complement operator. Expansion in complete homogeneous symmetric polynomials is then diagonalized by a Dirichlet average, converted to a beta integral, and smoothed by a Laplace transform into expectations of a shifted gamma random variable. The remaining positivity follows from a uniform odd-to-even gamma moment inequality.

For $1/2 < p < 2/3$, the Hilbert–Schmidt budget allows at most one compression eigenvalue above $q = 1 - p$. Isolating this frontier eigenvalue and retaining both its direct coupling and the coupling forced into the safe spectral subspace leads to a one-dimensional Huber-type envelope. Two canonical dual witnesses settle its quadratic and linear branches. The only finite residue is a pair of exact univariate Bernstein certificates at $m = 9$. The cycles C_3, C_5, C_7 are also given separate elementary proofs.

A final section extracts the reusable mechanisms: one-step Jacobi coefficient striping, first-return and cyclic excursion expansions, Dirichlet probabilistic polarization, beta–gamma/Laplace positivity transfer, the Laguerre differential equation hidden in the gamma moment lemma, and the one-dangerous-mode/Huber principle. It also explains why the same threshold $q = 1/3$ emerges independently from the two sides.

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1 The theorem, notation, and proof map

For a finite graph F and a graphon W , write

$$t(F, W) = \int_{[0,1]^{V(F)}} \prod_{ij \in E(F)} W(x_i, x_j) \prod_{i \in V(F)} dx_i.$$

In particular,

$$p = t(K_2, W) = \int_{[0,1]^2} W(x, y) \, dx \, dy$$

is the edge density. We prove the following theorem.

Theorem 1.1 (Odd-cycle Goodman bound). *Let W be a graphon of edge density $p \in [0, 1]$. For every odd integer $m \geq 3$,*

$$t(C_m, W) \geq p^m - p(1-p)^{m-1}. \quad (1)$$

Put

$$U = 1 - W, \quad q = t(K_2, U) = 1 - p.$$

If $p \leq 1/2$, then

$$p^m - pq^{m-1} = p(p^{m-1} - q^{m-1}) \leq 0,$$

whereas $t(C_m, W) \geq 0$. Thus only $p > 1/2$, or $q < 1/2$, needs proof. The nontrivial range splits into

$$\underbrace{0 \leq q \leq \frac{1}{3}}_{p \geq 2/3} \quad \text{and} \quad \underbrace{\frac{1}{3} < q < \frac{1}{2}}_{1/2 < p < 2/3}.$$

1.1 Proof map

The argument has four modules.

Module 1. Section 2 gives elementary proofs for $m = 3, 5, 7$, at every edge density.

Module 2. Sections 3 and 4 proves all odd lengths when $q \leq 1/3$. The principal chain is

$$\begin{aligned} &\text{Schur cancellation} \longrightarrow \text{Dirichlet diagonalization} \\ &\longrightarrow \text{beta integral} \longrightarrow \text{gamma smoothing.} \end{aligned}$$

Module 3. Sections 5 to 9 proves all odd $m \geq 9$ when $1/3 < q < 1/2$. The principal chain is

$$\begin{aligned} &\text{one frontier eigenvalue} \longrightarrow \text{two coupling channels} \\ &\longrightarrow \text{Huber duality} \longrightarrow \text{two scalar witnesses.} \end{aligned}$$

Module 4. Section 11 isolates general principles and connects them to Jacobi matrices, orthogonal polynomials, continued fractions, lattice paths, Stein identities, and Laguerre polynomials.

The final assembly is in Section 10. No numerical optimization or interval covering is used. The two fixed Bernstein coefficient lists in Section A are exact rational certificates.

1.2 Step-graphon reduction

Lemma 1.2 (Finite-rank reduction). *For each fixed odd $m \geq 3$, it is enough to prove (1) for symmetric step graphons.*

Proof. Let $(\mathcal{P}_n)$ be refining finite measurable partitions generating the Borel sigma-algebra, and put

$$W_n = \mathbb{E}[W \mid \mathcal{P}_n \times \mathcal{P}_n].$$

Then W_n is a symmetric step graphon, $0 \leq W_n \leq 1$, and $W_n \rightarrow W$ in $L^2([0, 1]^2)$. Conditional expectation preserves the integral, so $t(K_2, W_n) = p$ exactly. The associated integral operators converge in Hilbert–Schmidt norm. A telescoping expansion of $T_{W_n}^m - T_W^m$, followed by Schatten–Hölder, gives

$$\text{Tr}(T_{W_n}^m) \longrightarrow \text{Tr}(T_W^m).$$

Since $t(C_m, W) = \text{Tr}(T_W^m)$, the desired inequality passes to the limit. The graphon–operator dictionary used here is standard; see, for example, Lovász [10]. $\square$

Remark 1.3 (Formal versus analytic power series). *All coefficient extractions below may be read formally. After the finite-rank reduction, the relevant resolvents are rational functions analytic near the origin, so ordinary determinant and logarithm identities are also legitimate analytically.*

2 The elementary cases $m = 3, 5, 7$

Throughout this section, W is any graphon, $p = t(K_2, W)$,

$$U = 1 - W, \quad q = t(K_2, U) = 1 - p,$$

and $T = T_U$ is the integral operator of U . When $q > 1/2$, the right side of (1) is nonpositive, so the result is trivial. We may therefore assume

$$0 \leq q \leq \frac{1}{2}. \quad (2)$$

The arguments below actually prove the three cases at every edge density.

2.1 The triangle: Goodman's one-line argument

The following is the graphon form of Goodman's classical counting argument [7].

Proposition 2.1. *For every graphon W ,*

$$t(C_3, W) \geq p^3 - p(1 - p)^2.$$

Proof. The target polynomial is

$$p^3 - p(1 - p)^2 = 2p^2 - p.$$

Let $d_W(x) = \int_0^1 W(x, y) dy$. Since $ab \geq a + b - 1$ for $a, b \in [0, 1]$,

$$\int_0^1 W(x, z)W(y, z) dz \geq d_W(x) + d_W(y) - 1.$$

Multiplying by $W(x, y) \geq 0$, integrating in (x, y) , and applying Cauchy–Schwarz gives

$$\begin{aligned} t(K_3, W) &\geq \int W(x, y)(d_W(x) + d_W(y) - 1) dx dy \\ &= 2 \int_0^1 d_W(x)^2 dx - p \geq 2p^2 - p. \end{aligned}$$

Since $C_3 = K_3$, this is the claim. □

2.2 Path moments in the complement

Write

$$T\mathbf{1} = q\mathbf{1} + g, \quad g \perp \mathbf{1}. \quad (3)$$

For $j \geq 1$, let P_j denote the path with j edges and put

$$x_j = t(P_j, U) = \langle \mathbf{1}, T^j \mathbf{1} \rangle.$$

Thus $x_1 = q$. The following first moments will be used for C_5 .

Lemma 2.2. *With (3),*

$$\begin{aligned} x_2 &= q^2 + \|g\|^2, \\ x_3 &= q^3 + 2q\|g\|^2 + \langle g, Tg \rangle, \\ x_4 &= q^4 + 3q^2\|g\|^2 + 2q\langle g, Tg \rangle + \|Tg\|^2. \end{aligned}$$

Proof. The first identity is $x_2 = \|T\mathbf{1}\|^2$. Next, $T^2\mathbf{1} = q^2\mathbf{1} + qg + Tg$, while

$$\langle \mathbf{1}, Tg \rangle = \langle T\mathbf{1}, g \rangle = \|g\|^2.$$

Expanding $x_3 = \langle T\mathbf{1}, T^2\mathbf{1} \rangle$ and $x_4 = \|T^2\mathbf{1}\|^2$ gives the displayed formulas. $\square$

For C_7 , let $H_0 = \mathbf{1}^\perp$ and

$$A = P_{H_0}TP_{H_0}.$$

Define

$$s_j = \langle g, A^j g \rangle \quad (j \geq 0). \quad (4)$$

The spectral theorem supplies a finite positive measure μ on the real line such that

$$s_j = \int \lambda^j d\mu(\lambda), \quad s_0 = \|g\|^2 = \mu(\mathbb{R}).$$

A useful way to verify all path formulas is to write

$$T^n\mathbf{1} = a_n\mathbf{1} + h_n, \quad h_n \perp \mathbf{1}.$$

Then $a_n = x_n$, and the block form of T gives the recursion

$$a_{n+1} = qa_n + \langle g, h_n \rangle, \quad h_{n+1} = a_ng + Ah_n, \quad (a_0, h_0) = (1, 0). \quad (5)$$

Repeated use of (5) yields

$$\begin{aligned} x_2 &= q^2 + s_0, \\ x_3 &= q^3 + 2qs_0 + s_1, \\ x_4 &= q^4 + 3q^2s_0 + 2qs_1 + s_0^2 + s_2, \\ x_5 &= q^5 + 4q^3s_0 + 3q^2s_1 + 3qs_0^2 + 2qs_2 + 2s_0s_1 + s_3, \\ x_6 &= q^6 + 5q^4s_0 + 4q^3s_1 + 6q^2s_0^2 + 3q^2s_2 + 6qs_0s_1 + 2qs_3 \\ &\quad + s_0^3 + 2s_0s_2 + s_1^2 + s_4. \end{aligned} \quad (6)$$

2.3 The pentagon: one completion of a square

Proposition 2.3. *For every graphon W ,*

$$t(C_5, W) \geq p^5 - p(1-p)^4.$$

Proof. Assume (2). Expand the five factors $1 - U_e$ around the cycle. Every proper selected subgraph is a disjoint union of paths, and homomorphism density is multiplicative over disjoint unions. The relevant counts are

number of selected edges	component lengths	multiplicity
0	—	1
1	1	5
2	2	5
2	1 + 1	5
3	3	5
3	2 + 1	5
4	4	5.

Consequently, with $c_5 = t(C_5, U)$,

$$t(C_5, 1 - U) = 1 - 5q + 5x_2 + 5q^2 - 5x_3 - 5qx_2 + 5x_4 - c_5.$$

Since $0 \leq U \leq 1$, deleting one edge from the cycle integrand gives $c_5 \leq x_4$. Therefore

$$t(C_5, W) \geq 1 - 5q + 5q^2 + 5(1 - q)x_2 - 5x_3 + 4x_4. \quad (7)$$

The target is

$$(1 - q)^5 - (1 - q)q^4 = 1 - 5q + 10q^2 - 10q^3 + 4q^4.$$

Subtracting it from (7) and using [Theorem 2.2](#) gives

$$\begin{aligned} \Phi_5 &= 4 \|Tg\|^2 + (8q - 5) \langle g, Tg \rangle + (12q^2 - 15q + 5) \|g\|^2 \\ &= 4 \left\| Tg + \frac{8q - 5}{8} g \right\|^2 + \left(8q^2 - 10q + \frac{55}{16} \right) \|g\|^2. \end{aligned} \quad (8)$$

The residual quadratic has discriminant

$$100 - 4 \cdot 8 \cdot \frac{55}{16} = -10 < 0$$

and positive leading coefficient. Thus $\Phi_5 \geq 0$, proving the claim. $\square$

2.4 The heptagon: a univariate sum of squares

The final certificate below is an explicit one-variable sum of squares; for general background on positivity of polynomials on an interval, see Powers–Reznick [\[11\]](#).

Proposition 2.4. *For every graphon W ,*

$$t(C_7, W) \geq p^7 - p(1 - p)^6.$$

Proof. Assume (2). The proper edge subsets of a labelled 7-cycle have the following run-length types.

k	types and multiplicities for k selected edges
0	1
1	7(1)
2	7(2) + 14(1 + 1)
3	7(3) + 21(2 + 1) + 7(1 + 1 + 1)
4	7(4) + 14(3 + 1) + 7(2 + 2) + 7(2 + 1 + 1)
5	7(5) + 7(4 + 1) + 7(3 + 2)
6	7(6).

Here, for example, $3 + 1$ denotes a disjoint union of a three-edge path and an edge. Expanding $t(C_7, 1 - U)$, and using $c_7 = t(C_7, U) \leq x_6$, gives

$$\begin{aligned} t(C_7, W) &\geq 6x_6 - 7x_5 + 7(1 - q)x_4 + 7(2q - 1)x_3 \\ &\quad + 7(q^2 - 3q + 1)x_2 + 7x_2^2 - 7x_2x_3 \\ &\quad + 1 - 7q + 14q^2 - 7q^3. \end{aligned} \quad (9)$$

Since

$$(1 - q)^7 - (1 - q)q^6 = 1 - 7q + 21q^2 - 35q^3 + 35q^4 - 21q^5 + 6q^6,$$

substitution of (6) reduces the desired inequality to

$$\begin{aligned}\Phi_7 = & 6s_4 + (12q - 7)s_3 + (18q^2 - 21q + 7)s_2 \\ & + (24q^3 - 42q^2 + 28q - 7)s_1 \\ & + (30q^4 - 70q^3 + 70q^2 - 35q + 7)s_0 \\ & + 12s_0s_2 + (36q - 21)s_0s_1 + (36q^2 - 42q + 14)s_0^2 + 6s_0^3 + 6s_1^2.\end{aligned}\tag{10}$$

Put $r = s_0 = \mu(\mathbb{R})$. Grouping the terms integrated against μ ,

$$\Phi_7 = 6s_1^2 + \int R_{q,r}(\lambda) d\mu(\lambda),\tag{11}$$

where

$$\begin{aligned}R_{q,r}(\lambda) = & P_q(\lambda) + rB_q(\lambda) + 6r^2, \\ P_q(\lambda) = & 6\lambda^4 + (12q - 7)\lambda^3 + (18q^2 - 21q + 7)\lambda^2 \\ & + (24q^3 - 42q^2 + 28q - 7)\lambda + 30q^4 - 70q^3 + 70q^2 - 35q + 7, \\ B_q(\lambda) = & 12\lambda^2 + (36q - 21)\lambda + 36q^2 - 42q + 14.\end{aligned}$$

We show that both P_q and B_q are nonnegative.

The minimum of B_q as a quadratic in λ occurs at $\lambda_* = (7 - 12q)/8$, and

$$B_q(\lambda_*) = \frac{144q^2 - 168q + 77}{16} \geq \frac{29}{16},$$

because

$$144q^2 - 168q + 48 = 144 \left(q - \frac{1}{2} \right) \left(q - \frac{2}{3} \right) \geq 0$$

on $[0, 1/2]$.

For P_q , set

$$\begin{aligned}D(q) = & 288q^2 - 336q + 119, \\ C(q) = & 24q^3 - 42q^2 + 28q - 7, \\ N(q) = & 5184q^6 - 18144q^5 + 28602q^4 - 25802q^3 \\ & + 13874q^2 - 4165q + 539.\end{aligned}$$

Direct expansion gives the exact identity

$$\begin{aligned}P_q(\lambda) = & 6 \left(\lambda^2 + \left(q - \frac{7}{12} \right) \lambda \right)^2 \\ & + \frac{D(q)}{24} \left(\lambda + \frac{12C(q)}{D(q)} \right)^2 + \frac{N(q)}{D(q)}.\end{aligned}\tag{12}$$

Writing $y = 1/2 - q \geq 0$,

$$D(q) = 288y^2 + 48y + 23 > 0$$

and

$$\begin{aligned}8N(q) = & 41472y^6 + 20736y^5 + 21456y^4 + 7984y^3 \\ & + 2032y^2 + 316y + 11 > 0.\end{aligned}$$

Thus $P_q(\lambda) \geq 0$ for every real λ . Since $r \geq 0$, $R_{q,r}(\lambda) \geq 0$, and (11) yields $\Phi_7 \geq 0$. $\square$

Remark 2.5. The C_5 and C_7 arguments use no positivity of the kernel operator T_U ; a pointwise nonnegative kernel need not define a positive semidefinite operator. Only self-adjointness, the positive spectral measure associated with a vector, and the displayed algebraic identities are used.

3 The common complement-operator framework

By [Theorem 1.2](#), determinant arguments may be carried out for a step graphon. Decompose

$$L^2([0, 1]) = \mathbb{R}\mathbf{1} \oplus \mathbf{1}^\perp$$

and write the complement operator as

$$T_U = \begin{pmatrix} q & g^* \\ g & A \end{pmatrix}, \quad g = T_U \mathbf{1} - q\mathbf{1}, \quad A = P_{\mathbf{1}^\perp} T_U P_{\mathbf{1}^\perp}. \quad (13)$$

The vector $\mathbf{1}$ has norm one because the underlying probability space has total mass one.

Lemma 3.1 (Square budget). *One has*

$$\mathrm{Tr}(A^2) + 2\|g\|^2 \leq pq. \quad (14)$$

Proof. Since $0 \leq U \leq 1$,

$$\mathrm{Tr}(T_U^2) = \int_{[0,1]^2} U(x, y)^2 \, dx \, dy \leq \int_{[0,1]^2} U(x, y) \, dx \, dy = q.$$

On the other hand, (13) gives

$$\mathrm{Tr}(T_U^2) = q^2 + 2\|g\|^2 + \mathrm{Tr}(A^2).$$

Subtracting q^2 proves the claim. $\square$

Lemma 3.2 (Half-norm lemma). *The spectrum of A lies in $[-1/2, 1/2]$; equivalently,*

$$\|A\|_{\mathrm{op}} \leq \frac{1}{2}. \quad (15)$$

Proof. Let $f \perp \mathbf{1}$ and $\|f\| = 1$, and write $f = f_+ - f_-$. Since $\int f = 0$,

$$\int f_+ = \int f_- =: a, \quad 2a = \|f\|_1 \leq \|f\|_2 = 1.$$

Using $0 \leq U \leq 1$,

$$\langle f, T_U f \rangle \leq \left(\int f_+ \right)^2 + \left(\int f_- \right)^2 = 2a^2 \leq \frac{1}{2},$$

and similarly

$$\langle f, T_U f \rangle \geq -2 \left(\int f_+ \right) \left(\int f_- \right) = -2a^2 \geq -\frac{1}{2}.$$

The variational characterization of the norm of a self-adjoint operator proves the result. $\square$

Remark 3.3 (No positivity-of-operator assumption). *The pointwise inequality $U \geq 0$ does not imply that T_U or its compression A is positive semidefinite. The proof above controls the quadratic form directly and is the only spectral support statement needed.*

Let E_A be the spectral resolution of A , and define the finite positive measure

$$\mu(B) = \langle g, E_A(B)g \rangle. \quad (16)$$

Then μ is supported on $[-1/2, 1/2]$, and

$$\langle g, (I - zA)^{-1}g \rangle = \int_{-1/2}^{1/2} \frac{d\mu(\lambda)}{1 - \lambda z}. \quad (17)$$

This positive spectral measure is the bridge from the graphon operator to one-variable moment inequalities.

4 The dense region $p \geq 2/3$: Dirichlet and gamma smoothing

Throughout this section,

$$0 \leq q \leq \frac{1}{3}, \quad p = 1 - q, \quad (18)$$

and $m \geq 3$ is odd. We prove (1) for all such m . The proof is first presented as a sequence of exact reductions; a conceptual interpretation is given in [Section 11](#).

4.1 A two-sided spectral shift and cancellation of the tail

Define

$$R_-(z) = \langle g, (I + zA)^{-1}g \rangle = \int \frac{d\mu(\lambda)}{1 + \lambda z}, \quad (19)$$

$$R_+(z) = \langle g, (I - zA)^{-1}g \rangle = \int \frac{d\mu(\lambda)}{1 - \lambda z}, \quad (20)$$

and

$$L_-(z) = -\log \left(1 - \frac{z^2 R_-(z)}{1 - pz} \right), \quad (21)$$

$$L_+(z) = -\log \left(1 - \frac{z^2 R_+(z)}{1 - qz} \right). \quad (22)$$

For odd m , put

$$S_m = m[z^m](L_-(z) + L_+(z)). \quad (23)$$

Lemma 4.1 (Two-sided odd cancellation). *For every odd $m \geq 3$,*

$$t(C_m, W) + t(C_m, U) = p^m + q^m + S_m. \quad (24)$$

Consequently,

$$t(C_m, W) - (p^m - pq^{m-1}) = q^{m-1} + S_m - t(C_m, U). \quad (25)$$

Proof. The operator of $W = 1 - U$ is

$$T_W = \begin{pmatrix} p & -g^* \\ -g & -A \end{pmatrix}.$$

Conjugating by the sign flip on $\mathbf{1}^\perp$, it is unitarily equivalent to

$$M = \begin{pmatrix} p & g^* \\ g & -A \end{pmatrix}.$$

Schur complementation gives

$$\begin{aligned} \det(I - zM) &= \det(I + zA)(1 - pz) \left(1 - \frac{z^2 R_-(z)}{1 - pz} \right), \\ \det(I - zT_U) &= \det(I - zA)(1 - qz) \left(1 - \frac{z^2 R_+(z)}{1 - qz} \right). \end{aligned}$$

Using

$$-\log \det(I - zX) = \sum_{j \geq 1} \frac{\text{Tr}(X^j)}{j} z^j,$$

extract the coefficient of z^m and add the two formulas. Because m is odd, the contributions of $\det(I + zA)$ and $\det(I - zA)$ cancel. The scalar factors contribute $p^m + q^m$, and the remaining factors contribute S_m . This proves (24). Finally, $q^m + pq^{m-1} = q^{m-1}$, giving (25). $\square$

The cancellation is the decisive reason for using *both* W and its complement in this region: all odd moments $\text{Tr}(A^m)$ of the inaccessible compressed tail disappear. For background on determinant/trace expansions and finite-rank perturbations, see Simon [12].

4.2 Replacing a complement cycle by a path

Let P_j denote the path with j edges and set

$$x_j = t(P_j, U) = \langle \mathbf{1}, T_U^j \mathbf{1} \rangle. \quad (26)$$

Deleting one factor from the integrand of $t(C_m, U)$, and using $0 \leq U \leq 1$, gives

$$t(C_m, U) \leq x_{m-1}. \quad (27)$$

Block inversion in (13) gives the moment generating function

$$\sum_{j \geq 0} x_j z^j = \langle \mathbf{1}, (I - zT_U)^{-1} \mathbf{1} \rangle = \frac{1}{1 - qz - z^2 R_+(z)}. \quad (28)$$

Thus it is enough to prove

$$\Phi_m := q^{m-1} + S_m - x_{m-1} \geq 0. \quad (29)$$

4.3 Expansion in complete homogeneous polynomials

For variables $y_1, \dots, y_k$, define

$$h_d(y_1, \dots, y_k) = \sum_{\substack{\alpha_1, \dots, \alpha_k \geq 0 \\ \alpha_1 + \dots + \alpha_k = d}} y_1^{\alpha_1} \cdots y_k^{\alpha_k}, \quad h_0 = 1.$$

Equivalently,

$$\prod_{i=1}^k \frac{1}{1 - y_i z} = \sum_{d \geq 0} h_d(y_1, \dots, y_k) z^d. \quad (30)$$

The notation $a^{\times r}$ means that a is repeated r times.

Proposition 4.2 (Exact coefficient expansion). *For odd $m \geq 3$,*

$$\Phi_m = \sum_{r=1}^{(m-1)/2} \int \cdots \int P_{m,r}(q; \lambda_1, \dots, \lambda_r) \prod_{i=1}^r d\mu(\lambda_i), \quad (31)$$

where $n = m - 2r$ and

$$\begin{aligned} P_{m,r}(q; \lambda_1, \dots, \lambda_r) = & \frac{m}{r} \left[h_n(p^{\times r}, -\lambda_1, \dots, -\lambda_r) + h_n(q^{\times r}, \lambda_1, \dots, \lambda_r) \right] \\ & - h_{n-1}(q^{\times(r+1)}, \lambda_1, \dots, \lambda_r). \end{aligned} \quad (32)$$

Proof. Put

$$Y_-(z) = \frac{z^2 R_-(z)}{1 - pz}, \quad Y_+(z) = \frac{z^2 R_+(z)}{1 - qz}.$$

From $-\log(1 - Y) = \sum_{r \geq 1} Y^r / r$,

$$m[z^m] L_-(z) = \sum_{r \geq 1} \frac{m}{r} [z^{m-2r}] \frac{R_-(z)^r}{(1 - pz)^r}.$$

The spectral representation gives

$$R_-(z)^r = \int \cdots \int \prod_{i=1}^r \frac{1}{1 + \lambda_i z} \prod_{i=1}^r d\mu(\lambda_i).$$

By (30), the r -th contribution is the integral of

$$\frac{m}{r} h_n(p^{\times r}, -\lambda_1, \dots, -\lambda_r).$$

The same argument for L_+ produces the second term in brackets in (32).

On the other hand, (28) expands as

$$\frac{1}{1 - qz - z^2 R_+(z)} = \sum_{r \geq 0} \frac{(z^2 R_+(z))^r}{(1 - qz)^{r+1}}.$$

The term $r = 0$ contributes q^{m-1} to x_{m-1} , cancelling the first term in (29). For $r \geq 1$, the coefficient of z^{m-1} is the integral of

$$h_{n-1}(q^{\times(r+1)}, \lambda_1, \dots, \lambda_r).$$

Combining the contributions proves the proposition. $\square$

4.4 Dirichlet diagonalization: a probabilistic polarization

Define the diagonal polynomial

$$\tilde{P}_{m,r}(q, \ell) = P_{m,r}(q; \ell, \dots, \ell). \quad (33)$$

Lemma 4.3 (Dirichlet diagonalization). *Let $(\Theta_1, \dots, \Theta_r)$ be uniformly distributed on the simplex*

$$\Theta_i \geq 0, \quad \sum_{i=1}^r \Theta_i = 1.$$

Then

$$P_{m,r}(q; \lambda_1, \dots, \lambda_r) = \mathbb{E} \tilde{P}_{m,r} \left(q, \sum_{i=1}^r \Theta_i \lambda_i \right). \quad (34)$$

Consequently,

$$\min_{\lambda_i \in [-1/2, 1/2]} P_{m,r}(q; \lambda_1, \dots, \lambda_r) = \min_{\ell \in [-1/2, 1/2]} \tilde{P}_{m,r}(q, \ell). \quad (35)$$

Proof. For every $j \geq 0$, the Dirichlet moment formula gives

$$\mathbb{E} \left(\sum_{i=1}^r \Theta_i \lambda_i \right)^j = \frac{h_j(\lambda_1, \dots, \lambda_r)}{\binom{j+r-1}{r-1}}. \quad (36)$$

Indeed, expand the power and use

$$\mathbb{E} \prod_{i=1}^r \Theta_i^{\alpha_i} = \frac{(r-1)! \prod_i \alpha_i!}{(j+r-1)!}, \quad \sum_i \alpha_i = j.$$

Also,

$$h_d(a^{\times k}, b_1, \dots, b_r) = \sum_{j=0}^d \binom{d-j+k-1}{k-1} a^{d-j} h_j(b_1, \dots, b_r). \quad (37)$$

Apply (37) to each term in (32) and then use (36); this gives (34). Every convex combination $\sum_i \Theta_i \lambda_i$ lies in $[-1/2, 1/2]$, so the expectation is at least the diagonal minimum. Conversely, taking $\lambda_1 = \dots = \lambda_r = \ell$ realizes every diagonal value. $\square$

Thus (31) will be nonnegative once we prove

$$\tilde{P}_{m,r}(q, \ell) \geq 0 \quad (38)$$

for

$$0 \leq q \leq \frac{1}{3}, \quad 1 \leq r \leq \frac{m-1}{2}, \quad -\frac{1}{2} \leq \ell \leq \frac{1}{2}.$$

This is the first major dimension reduction: an r -variable spectral polynomial is replaced exactly, not approximately, by one scalar variable.

4.5 A beta representation

Fix $r \geq 1$ and put

$$n = m - 2r. \quad (39)$$

Then $n \geq 1$ is odd. Let $\Xi \sim \text{Beta}(r, r)$, and for $x \in [0, 1]$ set

$$V_x = qx + \ell(1 - x), \quad W_x = px - \ell(1 - x). \quad (40)$$

Thus $V_x + W_x = x$. Define

$$\rho_{n,m}(u) = \frac{m}{n}(u^n + (1 - u)^n) - u^{n-1}. \quad (41)$$

Lemma 4.4 (Beta formula). *One has*

$$\tilde{P}_{m,r}(q, \ell) = \binom{n+2r-1}{2r-1} \frac{n}{r} \left[\frac{m}{n} \mathbb{E}(V_{\Xi}^n + W_{\Xi}^n) - \mathbb{E}(\Xi V_{\Xi}^{n-1}) \right] \quad (42)$$

$$= \binom{n+2r-1}{2r-1} \frac{n}{r} \mathbb{E} \left[\Xi^n \rho_{n,m} \left(q + \ell \frac{1 - \Xi}{\Xi} \right) \right]. \quad (43)$$

If $\ell > 0$, then

$$\tilde{P}_{m,r}(q, \ell) = \frac{\Gamma(m)}{r\Gamma(n)\Gamma(r)^2} \ell^{n+r} \int_0^\infty \frac{s^{r-1}}{(\ell + s)^m} \rho_{n,m}(q + s) \, ds. \quad (44)$$

Proof. For arbitrary a, b , comparison of coefficients with the beta moments shows

$$h_n(a^{\times r}, b^{\times r}) = \binom{n+2r-1}{2r-1} \mathbb{E}(a\Xi + b(1 - \Xi))^n. \quad (45)$$

Differentiating with respect to the common variable a , and noting that the derivative of the left side is $rh_{n-1}(a^{\times(r+1)}, b^{\times r})$, gives

$$h_{n-1}(a^{\times(r+1)}, b^{\times r}) = \binom{n+2r-1}{2r-1} \frac{n}{r} \mathbb{E} \left[\Xi (a\Xi + b(1 - \Xi))^{n-1} \right]. \quad (46)$$

Apply these formulas to the three terms of (32); the two linear forms are precisely V_{Ξ} and W_{Ξ} . This proves (42). If

$$u = q + \ell \frac{1 - x}{x},$$

then $V_x = xu$ and $W_x = x(1 - u)$, giving (43).

For $\ell > 0$, use the beta density

$$\frac{\Gamma(2r)}{\Gamma(r)^2} x^{r-1} (1 - x)^{r-1} \, dx$$

and make the projective substitution

$$s = \ell \frac{1 - x}{x}, \quad x = \frac{\ell}{\ell + s}.$$

Then

$$x^{n+r-1} (1 - x)^{r-1} \, dx = -\ell^{n+r} \frac{s^{r-1}}{(\ell + s)^m} \, ds.$$

Substitution into (43), together with

$$\binom{m-1}{2r-1} \frac{n}{r} \frac{\Gamma(2r)}{\Gamma(r)^2} = \frac{\Gamma(m)}{r\Gamma(n)\Gamma(r)^2},$$

proves (44). □

4.6 The nonpositive diagonal is pointwise safe

Lemma 4.5. *If $0 \leq q \leq 1/2$ and $\ell \leq 0$, then $\tilde{P}_{m,r}(q, \ell) \geq 0$.*

Proof. The identity

$$\rho_{n,m}(u) = \frac{m}{n}(1-u)((1-u)^{n-1} - u^{n-1}) + \left(\frac{m}{n} - 1\right)u^{n-1} \quad (47)$$

shows that $\rho_{n,m}(u) \geq 0$ whenever $u \leq 1/2$: the exponent $n-1$ is even and $|1-u| \geq |u|$. If $\ell \leq 0$, then for $x > 0$,

$$q + \ell \frac{1-x}{x} \leq q \leq \frac{1}{2}.$$

The integrand in (43) is therefore nonnegative. $\square$

Only $\ell > 0$ remains. Pointwise positivity of $\rho_{n,m}(q+s)$ fails for large s , so the beta integral must be used as a genuinely averaged object.

4.7 Laplace–gamma smoothing

Let $Y \sim \text{Gamma}(r, 1)$, with density

$$\frac{1}{\Gamma(r)} y^{r-1} e^{-y} \mathbf{1}_{(0,\infty)}(y).$$

Lemma 4.6 (Gamma smoothing). *If $\ell > 0$, then*

$$\tilde{P}_{m,r}(q, \ell) = \frac{\ell^{n+r}}{r\Gamma(n)\Gamma(r)} \int_0^\infty t^{n+r-1} e^{-\ell t} \mathbb{E} \rho_{n,m} \left(q + \frac{Y}{t} \right) dt. \quad (48)$$

Proof. Insert the Laplace representation

$$\frac{1}{(\ell+s)^m} = \frac{1}{\Gamma(m)} \int_0^\infty t^{m-1} e^{-t(\ell+s)} dt$$

into (44). The exchange of integrals is justified by absolute convergence: because n is odd, $\rho_{n,m}$ has degree at most $n-1$, while $m = n+2r$. With $y = ts$,

$$\int_0^\infty s^{r-1} e^{-ts} \rho_{n,m}(q+s) ds = \Gamma(r) t^{-r} \mathbb{E} \rho_{n,m} \left(q + \frac{Y}{t} \right).$$

Substitution gives (48). $\square$

The dependence on the diagonal parameter ℓ now occurs only through a positive weight. It remains to prove a statement independent of ℓ .

4.8 The shifted-gamma moment inequality

Lemma 4.7 (Uniform odd-to-even gamma moment bound). *Let $r, j \geq 1$ be integers and $Y \sim \text{Gamma}(r, 1)$. For every $z \geq 0$,*

$$3j \mathbb{E}(zY - 1)^{2j-1} \leq (r+j) \mathbb{E}(zY - 1)^{2j}. \quad (49)$$

We postpone its proof for one subsection and first show exactly how it finishes the dense region.

Proposition 4.8 (Shifted-gamma positivity). *Let $n \geq 1$ be odd, $r \geq 1$, and $m = n+2r$. If $0 \leq q \leq 1/3$, then for every $x \geq 0$,*

$$\mathbb{E} \rho_{n,m}(q + xY) \geq 0, \quad Y \sim \text{Gamma}(r, 1). \quad (50)$$

Proof. Set

$$A_n(u) = u^n + (1 - u)^n.$$

Since $m = n + 2r$, direct differentiation gives

$$n\rho_{n,m}(u) = 2rA_n(u) - (1 - u)A'_n(u). \quad (51)$$

Because n is odd,

$$A_n\left(\frac{1}{2} + v\right) = \sum_{j=0}^{(n-1)/2} a_j v^{2j}, \quad a_j = 2^{2j+1-n} \binom{n}{2j} > 0. \quad (52)$$

Substituting into (51) yields

$$n\rho_{n,m}\left(\frac{1}{2} + v\right) = 2ra_0 + \sum_{j=1}^{(n-1)/2} a_j (2(r+j)v^{2j} - jv^{2j-1}). \quad (53)$$

Put

$$\delta = \frac{1}{2} - q \geq \frac{1}{6}.$$

For fixed $x \geq 0$, write $z = x/\delta$. Then

$$q + xY - \frac{1}{2} = \delta(zY - 1).$$

By [Theorem 4.7](#), for each $j \geq 1$,

$$\begin{aligned} & \mathbb{E} \left[2(r+j)(\delta(zY - 1))^{2j} - j(\delta(zY - 1))^{2j-1} \right] \\ & \geq \delta^{2j-1}(r+j) \left(2\delta - \frac{1}{3} \right) \mathbb{E}(zY - 1)^{2j} \geq 0. \end{aligned}$$

Every coefficient a_j is positive, and the constant term in (53) is $2ra_0 > 0$. Taking expectations proves (50). $\square$

Remark 4.9 (The threshold is retained exactly). *The density assumption enters only through*

$$2\delta - \frac{1}{3} = 2\left(\frac{1}{3} - q\right).$$

Thus gamma smoothing reaches precisely $q = 1/3$; it does not merely yield a weaker artificial threshold.

4.9 Proof of the gamma moment inequality

Proof of Theorem 4.7. The case $z = 0$ is immediate. Assume $z > 0$, put $b = 1/z$, and define

$$F(b) = \mathbb{E}(Y - b)^{2j}.$$

Since $F'(b) = -2j\mathbb{E}(Y - b)^{2j-1}$, inequality (49) is equivalent to

$$H(b) := (r+j)F(b) + \frac{3b}{2}F'(b) \geq 0. \quad (54)$$

For $I_k(b) = \mathbb{E}(Y - b)^k$, gamma integration by parts gives

$$I_{k+1}(b) = (r+k-b)I_k(b) + kbI_{k-1}(b). \quad (55)$$

Indeed, applying

$$\mathbb{E}[Y f'(Y)] = \mathbb{E}[(Y - r)f(Y)] \quad (56)$$

to $f(y) = (y - b)^k$ yields (55). Taking $k = 2j - 1$, and using $F' = -2jI_{2j-1}$ and $F'' = 2j(2j - 1)I_{2j-2}$, gives the differential equation

$$bF''(b) + (b - r - 2j + 1)F'(b) - 2jF(b) = 0. \quad (57)$$

At a zero of H , equations (54) and (57) give

$$\frac{H'(b)}{F(b)} = \frac{3(r + 4j)b - (r + j)(5r + 8j)}{3b}. \quad (58)$$

Set

$$b_* := \frac{(r + j)(5r + 8j)}{3(r + 4j)}. \quad (59)$$

Now $H(0) = (r + j)\mathbb{E}Y^{2j} > 0$, while

$$H(b) = (r + 4j)b^{2j} + O(b^{2j-1}) \quad (b \rightarrow \infty).$$

Formula (58) says that every zero below b_* is a strict downward crossing and every zero above b_* is a strict upward crossing. Therefore, if H were ever negative, its negative interval would contain b_* . It is enough to prove

$$H(b_*) > 0. \quad (60)$$

For the rest of the proof put $b = b_*$, and define

$$d = \frac{2(r + j)}{3}, \quad c = \frac{3jb}{r + j}. \quad (61)$$

Direct calculation gives

$$b - d = \frac{r(r + j)}{r + 4j} > 0, \quad cd = 2jb, \quad c - d = r + 2j - b. \quad (62)$$

Let

$$G(x) = x^{2j}(x + b)^r e^{-x}, \quad -b \leq x < \infty. \quad (63)$$

Using (62), its derivative factors as

$$G'(x) = x^{2j-1}(x + b)^{r-1}e^{-x}(c - x)(x + d). \quad (64)$$

After the change of variables $x = y - b$, equation (54) becomes

$$\begin{aligned} \frac{\Gamma(r)e^b}{r + j}H(b) &= \int_{-b}^{\infty} x^{2j-1}(x - c)(x + b)^{r-1}e^{-x} dx \\ &= - \int_{-b}^{\infty} \frac{G'(x)}{x + d} dx. \end{aligned} \quad (65)$$

The apparent singularity at $x = -d$ is removable because $G'(-d) = 0$. Write $G_0 = G(-d)$. Integrating by parts separately on the two sides of $-d$ gives

$$- \int_{-b}^{\infty} \frac{G'(x)}{x + d} dx = \frac{G_0}{b - d} - \int_{-b}^{\infty} \frac{G(x) - G_0}{(x + d)^2} dx. \quad (66)$$

Thus (60) follows if $G(-d)$ is the global maximum of G .

By (64), the two nonzero local maxima occur at $-d$ and c . Their ratio is

$$\frac{G(c)}{G(-d)} = \left(\frac{c}{d}\right)^{2j} \left(\frac{b+c}{b-d}\right)^r e^{-(c+d)}. \quad (67)$$

Put $t = j/r > 0$. The logarithm of (67), divided by r , is

$$\begin{aligned} L(t) &= 2t \log \left(\frac{3t(5+8t)}{2(1+t)(1+4t)} \right) + \log \left(\frac{(1+4t)(5+8t)}{3(1+t)} \right) \\ &\quad - \frac{32t^2 + 25t + 2}{3(1+4t)}. \end{aligned} \quad (68)$$

We prove

$$L(t) < 0 \quad (t > 0). \quad (69)$$

Let

$$A(t) = \frac{3t(5+8t)}{2(1+t)(1+4t)}.$$

Direct differentiation gives

$$L'(t) = 2 \log A(t) - \frac{(32t^2 + 16t - 7)(32t^2 + 25t + 2)}{3(t+1)(4t+1)^2(8t+5)}, \quad (70)$$

$$L''(t) = -\frac{(32t^2 + 25t + 2)(128t^4 + 224t^3 + 120t^2 - 28t - 25)}{t(t+1)^2(4t+1)^3(8t+5)^2}. \quad (71)$$

The quartic in (71) has exactly one positive root: its coefficient sequence has one sign change, its value at zero is negative, and its leading coefficient is positive. Hence L' first increases and then decreases. Also $L'(t) \rightarrow -\infty$ as $t \downarrow 0$. At $t = 1$, the bound

$$\log x \geq \frac{2(x-1)}{x+1} \quad (x \geq 1)$$

gives

$$L'(1) \geq \frac{76}{59} - \frac{2419}{1950} > 0.$$

At $t = 3/2$, concavity of $\log$, together with $\log 2 < 7/10$, gives

$$L' \left(\frac{3}{2} \right) < \frac{111}{70} - \frac{19847}{12495} < 0.$$

Thus the only interior local maximum of L lies in $(1, 3/2)$, while

$$\lim_{t \downarrow 0} L(t) = \log \frac{5}{3} - \frac{2}{3} < 0.$$

It remains to control L on $[1, 3/2]$. Put

$$B(t) = \frac{(1+4t)(5+8t)}{3(1+t)}.$$

By concavity of $\log$,

$$\log A \leq \log 2 + \frac{A-2}{2}, \quad \log B \leq \log 13 + \frac{B-13}{13}.$$

Using $\log 2 < 7/10$ and $\log 13 < 13/5$, exact simplification gives

$$L(t) < \frac{3(96t^3 - 191t^2 - 16t + 46)}{130(t+1)(4t+1)}. \quad (72)$$

The cubic numerator is negative on $[1, 3/2]$. Its derivative has exactly one zero in that interval; this critical point is a minimum, while the endpoint values are

$$-65, \quad -\frac{335}{4}.$$

This proves (69). The elementary logarithm bounds follow directly from the exponential series.

It follows from (67) that $G(c) < G(-d)$, so $G(-d)$ is the global maximum. Equation (66) gives $H(b_*) > 0$. The zero-crossing argument then proves $H(b) \geq 0$ for all $b > 0$, which is equivalent to (49). $\square$

4.10 Completion of the dense-region proof

Theorem 4.10 (Dense-region theorem). *If $p \geq 2/3$, then (1) holds for every odd $m \geq 3$.*

Proof. Equivalently, $q \leq 1/3$. By Theorems 4.2 and 4.3, it is enough to prove (38). If $\ell \leq 0$, this is Theorem 4.5. If $\ell > 0$, Theorem 4.8 makes every expectation in (48) nonnegative, and the remaining weight is positive. Hence $\tilde{P}_{m,r}(q, \ell) \geq 0$ for every admissible r, ℓ . Therefore every integrand in (31) is nonnegative, so $\Phi_m \geq 0$. Finally, (25), (27), and (29) give

$$\begin{aligned} t(C_m, W) - (p^m - pq^{m-1}) &= q^{m-1} + S_m - t(C_m, U) \\ &\geq q^{m-1} + S_m - x_{m-1} = \Phi_m \geq 0. \end{aligned}$$

The finite-rank reduction returns the conclusion to arbitrary graphons. $\square$

5 The intermediate region: the one-frontier reduction for $m \geq 9$

From now on,

$$\frac{1}{3} < q < \frac{1}{2}, \quad \frac{1}{2} < p < \frac{2}{3}, \quad (73)$$

and $m \geq 9$ is odd. We continue with the common block decomposition (13), the square budget (14), and the spectral bound (15). By Theorem 1.2, we may again work with a step graphon during determinant manipulations.

5.1 The one-sided spectral-shift identity

The graphon operator of $W = 1 - U$ has block form

$$T_W = \begin{pmatrix} p & -g^* \\ -g & -A \end{pmatrix}.$$

Conjugating by the sign flip on $\mathbf{1}^\perp$ does not change traces, so we may instead use

$$M = \begin{pmatrix} p & g^* \\ g & -A \end{pmatrix}.$$

Lemma 5.1 (One-sided shift). *For every odd $m \geq 3$,*

$$t(C_m, W) = p^m - \text{Tr}(A^m) + m[z^m]\mathcal{L}(z), \quad (74)$$

where

$$\mathcal{L}(z) = -\log \left(1 - \frac{z^2 \langle g, (I + zA)^{-1} g \rangle}{1 - pz} \right). \quad (75)$$

Moreover, every coefficient of $\mathcal{L}(z)$ is nonnegative.

Proof. Schur complementation gives

$$\begin{aligned}\det(I - zM) &= \det(I + zA) (1 - pz - z^2 \langle g, (I + zA)^{-1} g \rangle) \\ &= \det(I + zA) (1 - pz) \left(1 - \frac{z^2 \langle g, (I + zA)^{-1} g \rangle}{1 - pz} \right).\end{aligned}$$

Taking $-\log$, and using

$$-\log \det(I - zM) = \sum_{j \geq 1} \frac{z^j}{j} \operatorname{Tr}(M^j),$$

we obtain (74) by comparing the coefficient of odd degree m .

It remains to check coefficientwise nonnegativity. Let μ_g be the spectral measure of A associated with g . For every integer $r \geq 2$,

$$\begin{aligned}[z^r] \frac{z^2 \langle g, (I + zA)^{-1} g \rangle}{1 - pz} \\ = \int \frac{p^{r-1} - (-\lambda)^{r-1}}{p + \lambda} d\mu_g(\lambda).\end{aligned}\tag{76}$$

By Theorem 3.2, $|\lambda| \leq 1/2 < p$. Thus the denominator and numerator in (76) are positive. The inner series in (75) therefore has nonnegative coefficients, and so does $-\log(1 - X) = \sum_{k \geq 1} X^k/k$. $\square$

For odd m , the linear term of the logarithm supplies the useful lower bound

$$m[z^m] \mathcal{L}(z) \geq m \int k_m(\lambda) d\mu_g(\lambda), \quad k_m(\lambda) = \frac{p^{m-1} - \lambda^{m-1}}{p + \lambda}.\tag{77}$$

The function k_m is strictly decreasing on $[0, p)$. Indeed, with $n = m - 1$,

$$k'_m(\lambda) = -\frac{p^n + np\lambda^{n-1} + (n-1)\lambda^n}{(p + \lambda)^2} < 0.$$

Also, for $0 \leq s < p$,

$$k_m(-s) = \frac{p^{m-1} - s^{m-1}}{p - s} > \frac{p^{m-1} - s^{m-1}}{p + s} = k_m(s).$$

Consequently,

$$k_m(\lambda) \geq k_m(L) \quad \text{whenever } |\lambda| \leq L < p.\tag{78}$$

5.2 No frontier, and uniqueness of a frontier

Lemma 5.2 (No-frontier case). *If the largest eigenvalue of A is at most q , then (1) holds.*

Proof. For every eigenvalue λ of A ,

$$\lambda^m \leq q^{m-2} \lambda^2 :$$

this is clear for $0 \leq \lambda \leq q$, while the left side is nonpositive for $\lambda < 0$. Therefore, by Theorem 3.1,

$$\operatorname{Tr}(A^m) \leq q^{m-2} \operatorname{Tr}(A^2) \leq pq^{m-1}.$$

The last term in (74) is nonnegative, so

$$t(C_m, W) \geq p^m - pq^{m-1}.$$

$\square$

We henceforth assume that A has an eigenvalue

$$\alpha > q. \quad (79)$$

It is unique. Indeed, two such eigenvalues would contribute more than $2q^2$ to $\text{Tr}(A^2)$, but

$$2q^2 > q(1 - q) = pq$$

precisely because $q > 1/3$. Define

$$L = \sqrt{pq - \alpha^2}. \quad (80)$$

The square budget implies that every other eigenvalue has absolute value at most L . Moreover,

$$0 \leq L < q < \alpha < \frac{1}{2} < p. \quad (81)$$

The strict inequality $L > 0$ will follow from the forced-variance lemma below; no argument before that lemma requires it. For example,

$$L^2 < pq - q^2 = q(1 - 2q) < q^2$$

because $q > 1/3$, while $\alpha < 1/2$ follows from [Theorem 3.2](#).

5.3 The forced-variance ceiling

The next inequality restricts the admissible pair (q, α) .

Lemma 5.3 (Forced variance). *Let $\phi \perp \mathbf{1}$ be a unit eigenvector with $A\phi = \alpha\phi$. Then*

$$\|g\|^2 \geq \frac{\alpha(\alpha - q)^2}{2(1 - 2\alpha)}. \quad (82)$$

Consequently,

$$\alpha^2 + q\alpha - q \leq 0, \quad (83)$$

and hence

$$q < \alpha \leq r(q) := \frac{\sqrt{q^2 + 4q} - q}{2}. \quad (84)$$

Proof. Put $a_\phi = \int |\phi|$. Since $\int \phi = 0$, $\int \phi_+ = \int \phi_- = a_\phi/2$. As $0 \leq U \leq 1$,

$$\begin{aligned} \alpha &= \langle \phi, T_U \phi \rangle \leq \iint U(x, y) (\phi_+(x)\phi_+(y) + \phi_-(x)\phi_-(y)) \, dx \, dy \\ &\leq \frac{a_\phi^2}{2}. \end{aligned}$$

Thus $a_\phi^2 \geq 2\alpha$.

Write $|\phi| = a_\phi \mathbf{1} + h$, where $h \perp \mathbf{1}$ and $\|h\|^2 = 1 - a_\phi^2$. Since $U \geq 0$,

$$\langle |\phi|, T_U |\phi| \rangle \geq \langle \phi, T_U \phi \rangle = \alpha.$$

Since α is the largest eigenvalue of A ,

$$\begin{aligned} \alpha &\leq qa_\phi^2 + 2a_\phi \langle g, h \rangle + \langle h, Ah \rangle \\ &\leq qa_\phi^2 + 2a_\phi \|g\| \sqrt{1 - a_\phi^2} + \alpha(1 - a_\phi^2). \end{aligned}$$

Therefore

$$(\alpha - q)a_\phi \leq 2\|g\| \sqrt{1 - a_\phi^2}.$$

The function $t/(1-t)$ is increasing, and $a_\phi^2 \geq 2\alpha$; after squaring we obtain (82).

Combining (82) with $\alpha^2 + 2\|g\|^2 \leq pq$, and multiplying by $1 - 2\alpha > 0$, gives

$$(1 - \alpha - q)(\alpha^2 + q\alpha - q) \leq 0.$$

The first factor is positive by (81), so (83) follows. Solving the quadratic gives (84). Finally, because $\alpha > q$, the right side of (82) is strictly positive. Thus $g \neq 0$, and the square budget gives $\alpha^2 < pq$. Consequently $L > 0$. $\square$

5.4 The master defect inequality

Choose the sign of ϕ later as convenient, and decompose

$$g = \gamma\phi + g_s, \quad g_s \perp \phi. \quad (85)$$

For odd m , define

$$A_m = 2L^{m-2} + mk_m(\alpha), \quad (86)$$

$$B_m = 2L^{m-2} + mk_m(L), \quad (87)$$

$$R_m = \alpha^m + L^m - pq^{m-1}. \quad (88)$$

By (81), (78), and the strict decrease of k_m ,

$$0 < A_m < B_m. \quad (89)$$

Proposition 5.4 (Master defect). *One has*

$$t(C_m, W) - (p^m - pq^{m-1}) \geq -R_m + A_m\gamma^2 + B_m\|g_s\|^2. \quad (90)$$

Proof. Let (λ_i) be the safe eigenvalues, i.e. all eigenvalues of A other than α . Since $|\lambda_i| \leq L$ and m is odd,

$$\sum_i \lambda_i^m \leq L^{m-2} \sum_i \lambda_i^2.$$

The budget (14) and (85) give

$$\sum_i \lambda_i^2 \leq L^2 - 2\gamma^2 - 2\|g_s\|^2.$$

Hence

$$-\text{Tr}(A^m) \geq -\alpha^m - L^m + 2L^{m-2}(\gamma^2 + \|g_s\|^2). \quad (91)$$

By (77) and (78),

$$m[z^m]\mathcal{L}(z) \geq mk_m(\alpha)\gamma^2 + mk_m(L)\|g_s\|^2. \quad (92)$$

Combining (74), (91), and (92) yields (90). $\square$

6 Eliminating the frontier eigenfunction: a Huber envelope

The terminology refers to the same quadratic-versus-linear active-set geometry as Huber's robust loss [8]. The exact dual formula below is a compact convex-concave minimax interchange of the type developed by Sion [13]; we nevertheless prove the required special case directly.

The master defect still contains the coupling variables γ and g_s . We now eliminate them together with the shape of the frontier eigenfunction.

Choose the sign of ϕ so that

$$b := \langle |\phi|, \phi \rangle \geq 0.$$

Set

$$z = \left(\int |\phi| \right)^2, \quad k = |\phi| - \sqrt{z} \mathbf{1} - b\phi, \quad K = \|k\|^2. \quad (93)$$

Then

$$z + b^2 + K = 1, \quad z \geq 2\alpha. \quad (94)$$

6.1 The direct and safe channels

Lemma 6.1 (Direct channel). *The frontier coupling obeys*

$$\gamma \leq u_\gamma := \frac{z - 2\alpha - 2\alpha b}{2\sqrt{z}}. \quad (95)$$

Proof. Since $T_U \phi = \gamma \mathbf{1} + \alpha \phi$ and

$$T_U \phi \leq T_U \phi_+ \leq \int \phi_+ = \frac{\sqrt{z}}{2}$$

pointwise, pairing with ϕ_+ gives

$$\frac{\gamma \sqrt{z}}{2} + \frac{\alpha(1+b)}{2} \leq \frac{z}{4}.$$

This is (95). □

Lemma 6.2 (Safe channel). *Put*

$$\mathcal{H} = \frac{(\alpha - q)z + (\alpha - L)K}{2\sqrt{z}}. \quad (96)$$

Then

$$b\gamma + \langle g_s, k \rangle \geq \mathcal{H}. \quad (97)$$

Consequently, if $K > 0$,

$$\|g_s\|^2 \geq \frac{(\mathcal{H} - b\gamma)_+^2}{K}. \quad (98)$$

Proof. Because $U \geq 0$,

$$\langle |\phi|, T_U |\phi| \rangle - \langle \phi, T_U \phi \rangle = 4 \langle \phi_+, T_U \phi_- \rangle \geq 0.$$

Expanding $|\phi| = \sqrt{z} \mathbf{1} + b\phi + k$, using $k \perp \mathbf{1}, \phi$, and using that every safe eigenvalue is at most L , we obtain

$$\begin{aligned} \alpha &\leq qz + 2\sqrt{z}(b\gamma + \langle g_s, k \rangle) + \alpha b^2 + \langle k, Ak \rangle \\ &\leq qz + 2\sqrt{z}(b\gamma + \langle g_s, k \rangle) + \alpha b^2 + LK. \end{aligned}$$

Since $z + b^2 + K = 1$, this rearranges to (97). If $\mathcal{H} - b\gamma > 0$, Cauchy–Schwarz gives $\|g_s\| \sqrt{K} \geq \mathcal{H} - b\gamma$; otherwise (98) is trivial. □

For fixed shape variables, [Theorem 5.4](#) and [Theorems 6.1](#) and [6.2](#) reduce the payment to

$$\mathcal{Q} = \min_{\gamma \leq u_\gamma} \left\{ A_m \gamma^2 + \frac{B_m}{K} (\mathcal{H} - b\gamma)_+^2 \right\}, \quad (99)$$

with the natural limiting interpretation when $K = 0$.

6.2 Exact shape elimination

Define

$$d = \alpha - q, \quad f = \alpha - L, \quad e = 1 - 2\alpha, \quad (100)$$

so $d, f, e > 0$. Put

$$C_m = \frac{B_m f \sqrt{2\alpha} e^2}{4\alpha^2}, \quad \xi = \frac{4\alpha^2 d}{e^2}, \quad \rho = \frac{A_m}{B_m} \frac{\sqrt{\alpha}}{2\sqrt{2} f}, \quad (101)$$

and define

$$\psi(\xi, \rho) = \min_{0 \leq v \leq 1} \left\{ \rho v^2 + (\xi - v + v^2)_+ \right\}. \quad (102)$$

Theorem 6.3 (Huber shape elimination). *For every admissible frontier configuration,*

$$A_m \gamma^2 + B_m \|g_s\|^2 \geq C_m \psi(\xi, \rho). \quad (103)$$

Consequently, (1) follows once

$$R_m \leq C_m \psi(\xi, \rho) \quad (104)$$

is proved for every pair (q, α) satisfying (73) and (84).

Proof. We first treat $K > 0$.

Suppose $u_\gamma \leq 0$. Every admissible γ is nonpositive, so $\mathcal{H} - b\gamma \geq \mathcal{H}$. Hence

$$\mathcal{Q} \geq B_m \frac{\mathcal{H}^2}{K}.$$

By (96),

$$\frac{\mathcal{H}^2}{K} = \frac{(dz + fK)^2}{4zK} \geq df,$$

because this inequality is equivalent to $(dz - fK)^2 \geq 0$. On the other hand, taking $v = 0$ in (102) gives

$$C_m \psi(\xi, \rho) \leq C_m \xi = B_m f d \sqrt{2\alpha} \leq B_m f d.$$

Thus (103) holds in this case.

Now suppose $u_\gamma > 0$. A minimizer in (99) may be taken with $\gamma \geq 0$: replacing a negative γ by 0 decreases both terms. The direct constraint gives

$$2\gamma\sqrt{z} + 2\alpha b \leq z - 2\alpha = e - b^2 - K \leq e - b^2.$$

Since $z \geq 2\alpha$,

$$b^2 + 2\alpha b + 2\gamma\sqrt{2\alpha} \leq e. \quad (105)$$

It follows that

$$0 \leq \gamma \leq \frac{e}{2\sqrt{2\alpha}}, \quad b \leq \beta(\gamma) := \sqrt{\alpha^2 + e - 2\gamma\sqrt{2\alpha}} - \alpha. \quad (106)$$

Rationalizing the square root,

$$\beta(\gamma) \leq \frac{e - 2\gamma\sqrt{2\alpha}}{2\alpha}. \quad (107)$$

Also, since $z \geq 2\alpha$ and $z \leq 1$,

$$\mathcal{H} \geq \frac{d\sqrt{2\alpha}}{2} + \frac{fK}{2}. \quad (108)$$

For every real w ,

$$\inf_{K>0} \frac{(w + fK/2)_+^2}{K} = 2f(w)_+. \quad (109)$$

Indeed, for $w \geq 0$ this is the arithmetic–geometric mean inequality, with equality at $K = 2w/f$; for $w < 0$, both sides vanish after choosing $0 < K \leq -2w/f$. Using (98), (108), and (109), and then maximizing b by (106), we obtain

$$\mathcal{Q} \geq \min_{0 \leq \gamma \leq e/(2\sqrt{2\alpha})} \left\{ A_m \gamma^2 + 2B_m f \left(\frac{d\sqrt{2\alpha}}{2} - \gamma\beta(\gamma) \right)_+ \right\}. \quad (110)$$

Set

$$v = \frac{2\gamma\sqrt{2\alpha}}{e} \in [0, 1].$$

By (107),

$$\gamma\beta(\gamma) \leq \frac{e^2 v(1-v)}{4\alpha\sqrt{2\alpha}}.$$

Therefore

$$\frac{d\sqrt{2\alpha}}{2} - \gamma\beta(\gamma) \geq \frac{\sqrt{2\alpha} e^2}{8\alpha^2} (\xi - v + v^2).$$

Furthermore,

$$A_m \gamma^2 = C_m \rho v^2, \quad 2B_m f \frac{\sqrt{2\alpha} e^2}{8\alpha^2} = C_m.$$

Substitution in (110) proves (103).

Finally suppose $K = 0$. The safe channel becomes

$$b\gamma \geq \frac{d\sqrt{z}}{2} \geq \frac{d\sqrt{2\alpha}}{2} > 0,$$

so $b, \gamma > 0$ and (105) still applies. With the same variable v ,

$$\frac{d\sqrt{2\alpha}}{2} \leq b\gamma \leq \frac{e^2 v(1-v)}{4\alpha\sqrt{2\alpha}},$$

which is equivalent to $\xi \leq v - v^2$. Thus the positive-part term in (102) vanishes at this v , and

$$A_m \gamma^2 = C_m \rho v^2 \geq C_m \psi(\xi, \rho).$$

This completes the proof. $\square$

6.3 The dual envelope and its two canonical witnesses

Proposition 6.4 (Dual form of the Huber envelope). *For $\xi \geq 0$ and $\rho > 0$,*

$$\psi(\xi, \rho) = \max_{0 \leq \lambda \leq 1} \left\{ \lambda \xi - \frac{\lambda^2}{4(\rho + \lambda)} \right\}. \quad (111)$$

Proof. Use

$$(x)_+ = \max_{0 \leq \lambda \leq 1} \lambda x.$$

The resulting function

$$(\rho + \lambda)v^2 - \lambda v + \lambda \xi$$

is convex in v , affine in λ , and the two domains are compact convex intervals. The elementary finite-dimensional minimax theorem (or Sion's theorem) permits the interchange of $\min_v$ and $\max_\lambda$. For fixed λ , the minimizer is

$$v = \frac{\lambda}{2(\rho + \lambda)} \in [0, 1/2],$$

and substitution gives (111). $\square$

Two choices of λ will be used throughout.

Corollary 6.5 (Quadratic and linear witnesses). *If $2\rho\xi \leq 1$, then*

$$C_m\psi(\xi, \rho) \geq \frac{2\alpha^3 A_m d^2}{e^2} \frac{1+4\xi}{1+2\xi}. \quad (112)$$

If $2\rho\xi > 1$, then

$$C_m\psi(\xi, \rho) > B_m f \sqrt{2\alpha} d \frac{1+4\xi}{2+4\xi}. \quad (113)$$

Proof. In the first case, take $\lambda = 2\rho\xi$ in (111). This gives

$$\psi(\xi, \rho) \geq \rho\xi^2 \frac{1+4\xi}{1+2\xi}.$$

A direct calculation from (101) gives

$$C_m \rho \xi^2 = \frac{2\alpha^3 A_m d^2}{e^2}.$$

In the second case, take $\lambda = 1$. Since $\rho > 1/(2\xi)$,

$$\frac{1}{4(1+\rho)} < \frac{\xi}{2+4\xi}.$$

Therefore

$$\psi(\xi, \rho) \geq \xi - \frac{1}{4(1+\rho)} > \xi \frac{1+4\xi}{2+4\xi}.$$

Finally,

$$C_m \xi = B_m f \sqrt{2\alpha} d.$$

□

7 The dimensionless chart

Set

$$N = m - 2. \quad (114)$$

Thus $N \geq 7$ is odd. Introduce

$$a = \frac{q}{\alpha}, \quad \ell = \frac{L}{\alpha}, \quad u = 1 - a, \quad D = 1 + a^2 + \ell^2, \quad \sigma = \frac{p}{\alpha}. \quad (115)$$

The relation $L^2 = pq - \alpha^2$ is equivalent to

$$\alpha = \frac{a}{D}, \quad q = \frac{a^2}{D}, \quad \sigma = \frac{1 + \ell^2}{a}. \quad (116)$$

Moreover,

$$d = \frac{au}{D}, \quad e = \frac{u^2 + \ell^2}{D}. \quad (117)$$

Lemma 7.1 (Chart domain). *The admissible parameters satisfy*

$$\begin{aligned} 0 < \ell < a < 1, \quad D < 3a^2, \quad \ell^2 \geq a(1-a) = au, \\ a > \frac{1 + \sqrt{13}}{6} > \frac{3}{4}, \quad 1 < \sigma < 2. \end{aligned} \quad (118)$$

Proof. The order $0 < L < q < \alpha$ gives $0 < \ell < a < 1$. Since $q = a^2/D > 1/3$, we have $D < 3a^2$. The ceiling $\alpha^2 + q\alpha - q \leq 0$, after substitution from (116), is exactly

$$\ell^2 \geq a - a^2 = au.$$

Hence $D \geq 1 + a$. Combining this with $D < 3a^2$ gives $3a^2 - a - 1 > 0$, which is the stated lower bound on a . Finally, $\sigma = p/\alpha > 1$ because $p > 1/2 > \alpha$, while $p < 2/3$ and $\alpha > q > 1/3$ give $\sigma < 2$. $\square$

Now put

$$\zeta = \frac{\ell^2}{u}, \quad v = \sigma - 1, \quad H_\zeta = \zeta + 1 + v. \quad (119)$$

Solving these relations gives

$$u = \frac{v}{H_\zeta}, \quad a = \frac{\zeta + 1}{H_\zeta}, \quad \ell^2 = \frac{\zeta v}{H_\zeta}. \quad (120)$$

The frontier ceiling $\ell^2 \geq au$ becomes

$$\zeta(\zeta + v) \geq 1. \quad (121)$$

Equivalently,

$$\zeta \geq \frac{\sqrt{v^2 + 4} - v}{2} \geq 1 - \frac{v}{2}. \quad (122)$$

In particular,

$$\zeta \geq a > \frac{3}{4}, \quad 0 < v < 1. \quad (123)$$

For later use, define

$$Q = v^2 + v\zeta + 2v + 2\zeta + 2. \quad (124)$$

A direct substitution gives

$$D = \frac{(\zeta + 1)Q}{H_\zeta^2}. \quad (125)$$

Since $D < 3a^2$,

$$Q < 3(\zeta + 1). \quad (126)$$

We shall repeatedly use the identity

$$\zeta H_\zeta^2 - (\zeta + 1)(\zeta + v)^2 = \zeta^2 + \zeta - v^2 > 0. \quad (127)$$

The positivity follows from (121):

$$\zeta^2 + \zeta - v^2 \geq 1 + \zeta(1 - v) - v^2 = (1 - v)(1 + v + \zeta) > 0.$$

7.1 The normalized defect and finite quotients

Normalize R_m by α^m :

$$F_N := \frac{R_m}{\alpha^m} = 1 - a^N - \ell^2(a^N - \ell^N). \quad (128)$$

Define

$$T_N = N - \zeta(a^N - \ell^N). \quad (129)$$

Since $1 - a^N \leq N(1 - a) = Nu$,

$$F_N \leq uT_N. \quad (130)$$

If $T_N \leq 0$, then $R_m \leq 0$, and the scalar target (104) is immediate. We may therefore assume

$$T_N > 0 \quad (131)$$

whenever convenient.

The normalized versions of A_m and B_m are

$$\frac{A_m}{\alpha^N} = 2\ell^N + (N+2)K_A, \quad K_A = \frac{\sigma^{N+1} - 1}{\sigma + 1}, \quad (132)$$

$$\frac{B_m}{\alpha^N} = 2\ell^N + (N+2)K_L, \quad K_L = \frac{\sigma^{N+1} - \ell^{N+1}}{\sigma + \ell}. \quad (133)$$

Because $N+1$ is even,

$$K_A = (\sigma - 1) \sum_{j=0}^{(N-1)/2} \sigma^{2j}, \quad (134)$$

$$K_L = (\sigma - \ell) \sum_{j=0}^{(N-1)/2} \sigma^{N-1-2j} \ell^{2j}. \quad (135)$$

7.2 Three elementary bounds for T_N

Lemma 7.2 (Defect bounds). *For every admissible chart point and every odd $N \geq 7$,*

$$T_N \leq N(1+v) - \zeta(1 - v^{N/2}), \quad (136)$$

$$T_N < N(1 - \ell^{N+1}). \quad (137)$$

If $T_N > 0$, then also

$$\zeta T_N < \frac{3N(N+1)}{2}(1+v). \quad (138)$$

Proof. By Bernoulli's inequality,

$$a^N \geq 1 - Nu.$$

Also $\ell^2 = \zeta v / H_\zeta < v$, so $\ell^N < v^{N/2}$. Since $\zeta u < v$,

$$\begin{aligned} T_N &= N - \zeta(a^N - \ell^N) \\ &\leq N - \zeta + N\zeta u + \zeta v^{N/2} \\ &\leq N(1+v) - \zeta(1 - v^{N/2}), \end{aligned}$$

proving (136).

Next, $D < 3a^2$ gives

$$\ell^2 < 2a^2 - 1 < (2a - 1)^2.$$

Thus $a - \ell > 1 - a = u$. By convexity of $x \mapsto x^N$,

$$a^N - \ell^N > N(a - \ell)\ell^{N-1} > Nu\ell^{N-1}.$$

Multiplication by $\zeta = \ell^2/u$ gives (137).

Assume now $T_N > 0$. From (136),

$$\zeta < Z := \frac{N(1+v)}{1 - v^{N/2}}. \quad (139)$$

Put $r = (N+1)/2$ and

$$g(t) = t \left[1 - \left(\frac{tv}{t+1+v} \right)^r \right].$$

If $\theta = (1 + v)/(t + 1 + v)$, then

$$g'(t) = 1 - [v(1 - \theta)]^r(1 + r\theta) \geq 0,$$

because $v^r \leq 1$ and Bernoulli's inequality gives $(1 - \theta)^{-r} \geq 1 + r\theta$. Therefore, using (137) and (139),

$$\zeta T_N < Ng(\zeta) < Ng(Z). \quad (140)$$

Let

$$\eta = v^{N/2}, \quad \omega = \frac{Nv}{N + 1 - \eta}.$$

A calculation gives

$$Ng(Z) = \frac{N^2(1 + v)}{1 - \eta} \left(1 - \omega^{(N+1)/2}\right).$$

As a function of $\eta \in (0, 1)$,

$$f(\eta) = \omega^{(N+1)/2} = \left(\frac{N}{N + 1 - \eta}\right)^{(N+1)/2} \eta^{(N+1)/N}$$

is convex. Indeed, with $r = (N + 1)/2$, $s = (N + 1)/N$, and $d = N + 1 - \eta$,

$$\frac{f''(\eta)}{f(\eta)} = \frac{r(r + 1)}{d^2} + \frac{2rs}{d\eta} + \frac{s(s - 1)}{\eta^2} > 0.$$

Moreover,

$$f(1) = 1, \quad f'(1) = \frac{3(N + 1)}{2N}.$$

The tangent inequality for a convex function therefore yields

$$1 - f(\eta) \leq \frac{3(N + 1)}{2N}(1 - \eta).$$

Substitution in (140) proves (138). □

8 The quadratic Huber branch

Assume

$$2\rho\xi \leq 1. \quad (141)$$

We shall prove (104) for every odd $N \geq 7$, hence for every odd $m \geq 9$.

By (112), discarding the factor $(1 + 4\xi)/(1 + 2\xi) > 1$, and using (132),

$$\frac{C_m \psi(\xi, \rho)}{\alpha^m} \geq 2(N + 2)\alpha \left(\frac{d}{e}\right)^2 K_A. \quad (142)$$

Since the summands in (134) have geometric mean $\sigma^{(N-1)/2}$, AM–GM gives

$$K_A \geq \frac{N + 1}{2} v \sigma^{(N-1)/2}. \quad (143)$$

Using (116), (117), and $v = u(1 + \zeta)/a$, the right side of (142) is at least

$$u(N + 2)(N + 1) \frac{a^2(1 + \zeta)}{D(u + \zeta)^2} \sigma^{(N-1)/2}. \quad (144)$$

Lemma 8.1 (Coefficient estimate). *One has*

$$\frac{a^2(1+\zeta)}{D(u+\zeta)^2} > \frac{1}{3\zeta}. \quad (145)$$

Proof. Using (120) and (125), the left side is

$$M = \frac{H_\zeta^2}{(\zeta+v)^2 Q}.$$

By (127),

$$\frac{H_\zeta^2}{(\zeta+v)^2} > \frac{\zeta+1}{\zeta},$$

while (126) gives $Q < 3(\zeta+1)$. Multiplication proves (145). $\square$

Combining (144) and (145),

$$\frac{C_m \psi(\xi, \rho)}{\alpha^m} > u \frac{(N+2)(N+1)}{3\zeta} (1+v)^{(N-1)/2}. \quad (146)$$

In view of (130), it remains to pay for uT_N .

Proposition 8.2 (Quadratic branch). *For every odd $N \geq 7$, assumption (141) implies*

$$R_m \leq C_m \psi(\xi, \rho).$$

Proof. If $T_N \leq 0$, the claim follows from (130). Assume $T_N > 0$.

First, (136) implies

$$\zeta T_N \leq N(1+v)\zeta - (1-v^{N/2})\zeta^2 \leq \frac{N^2(1+v)^2}{4(1-v^{N/2})}.$$

Thus (146) is sufficient whenever

$$E_N(v) := \frac{4(N+2)(N+1)}{3N^2} (1-v^{N/2})(1+v)^{(N-5)/2} \geq 1. \quad (147)$$

Second, the tangent estimate (138) shows that (146) is sufficient whenever

$$H_N(v) := \frac{2(N+2)}{9N} (1+v)^{(N-3)/2} \geq 1. \quad (148)$$

We now cover the whole interval $0 < v < 1$.

If $N \geq 9$ and $v \leq 3/5$, then

$$E_N(v) \geq \frac{4}{3} \left(1 - \left(\frac{3}{5} \right)^4 \right) = \frac{2176}{1875} > 1.$$

If $N \geq 9$ and $v \geq 3/5$, then

$$\frac{H_{N+2}(v)}{H_N(v)} = (1+v) \frac{N(N+4)}{(N+2)^2} > 1,$$

so $H_N(v)$ increases under $N \mapsto N+2$, and

$$H_9(3/5) = \frac{11264}{10125} > 1.$$

It remains to take $N = 7$. For $v \leq 9/10$,

$$E_7(v) \geq \frac{96}{49}(1 - v^3)(1 + v) \geq 1.$$

Indeed, $1 + v - v^3 - v^4$ is concave, so its minimum on $[0, 9/10]$ occurs at an endpoint; at $v = 9/10$ it equals $5149/10000 > 49/96$. For $v \geq 9/10$,

$$H_7(v) \geq \frac{2}{7} \left(\frac{19}{10} \right)^2 = \frac{361}{350} > 1.$$

Thus either (147) or (148) holds in every case, proving the proposition. $\square$

9 The linear Huber branch

Assume

$$2\rho\xi > 1. \tag{149}$$

Abbreviate

$$\varphi = \frac{1 + 4\xi}{2 + 4\xi}, \quad c_\xi = \sqrt{2\alpha} \varphi. \tag{150}$$

By (113), (133), and $f = \alpha(1 - \ell)$, $d = \alpha u$,

$$\frac{C_m \psi(\xi, \rho)}{\alpha^m} > u(N + 2)c_\xi(1 - \ell)K_L. \tag{151}$$

From (135),

$$K_L \geq (\sigma - \ell)\sigma^{N-1}. \tag{152}$$

In view of (130), it is therefore enough to prove

$$T_N \leq (N + 2)c_\xi(1 - \ell)(\sigma - \ell)\sigma^{N-1}. \tag{153}$$

9.1 The compensation lemma

The factor c_ξ may be small, but only where a short power of σ compensates for it.

Lemma 9.1 (Compensation). *On the whole admissible chart,*

$$c_\xi \sigma^{\zeta/4} \geq 1 - \ell^2. \tag{154}$$

Proof. We first show

$$\sqrt{2\alpha} \geq 1 - \ell^2. \tag{155}$$

Using $\alpha = a/D$, the square of (155) is

$$2a \geq (1 + a^2 + y)(1 - y)^2, \quad y = \ell^2.$$

The difference between the two sides has derivative

$$(1 - y)(1 + 2a^2 + 3y) > 0.$$

The smallest admissible y is $a(1 - a)$, where the difference is

$$(1 - a)(a^4 + a^2 + 2a - 1) > 0$$

because $a > 3/4$. This proves (155).

Next we show $\varphi\sigma^{\zeta/4} \geq 1$. From (101), (120), and (125),

$$\xi = \frac{4H_\zeta^2}{v(\zeta + v)^2Q}. \quad (156)$$

Equations (126) and (127) imply

$$4\zeta v\xi > \frac{16}{3}. \quad (157)$$

We claim that

$$\zeta v(1 + 4\xi) > 2(v + 2). \quad (158)$$

Indeed, the left side is larger than $\zeta v + 16/3$. If $\zeta \geq 2$, this is immediate. If $\zeta < 2$, then

$$v(2 - \zeta) < \frac{5}{4} < \frac{4}{3}$$

because $v < 1$ and $\zeta > 3/4$, which again proves (158).

The elementary inequality

$$\log(1 + v) \geq \frac{2v}{v + 2}$$

now gives

$$\frac{\zeta}{4} \log \sigma > \frac{1}{1 + 4\xi}.$$

Exponentiating and using $e^t > 1 + t$,

$$\sigma^{\zeta/4} > 1 + \frac{1}{1 + 4\xi} = \frac{2 + 4\xi}{1 + 4\xi} = \frac{1}{\varphi}.$$

Together with (155), this proves (154). $\square$

9.2 Two broad estimates for the linear payment

We first dispose of the range $\zeta \geq N$ when v is not large.

Lemma 9.2 (Above the cycle scale). *If $\zeta \geq N$, $0 < v \leq 5/8$, and $N \geq 7$ is odd, then (153) holds.*

Proof. By (136) and $\zeta \geq N$,

$$T_N \leq N(v + v^{N/2}). \quad (159)$$

Put $y = \sqrt{v}$. Since $\ell^2 < v$,

$$(1 - \ell)(\sigma - \ell) \geq (1 - y)(1 - y + y^2).$$

Also $q < \alpha$ and $p = (1 + v)\alpha$ imply $\alpha > 1/(2 + v)$, while $\varphi > 1/2$. Hence

$$c_\xi > \frac{1}{\sqrt{2(2 + v)}}. \quad (160)$$

It is enough to show

$$h(y) := \frac{(1 - y)(1 - y + y^2)(1 + y^2)^5}{y^2} > \sqrt{2(2 + y^2)}(1 + y^{N-2}). \quad (161)$$

Indeed, the omitted factors $(N + 2)/N$ and $(1 + y^2)^{N-6}$ are at least 1.

We claim that h is decreasing on $(0, 4/5]$. Differentiation gives

$$h'(y) = -\frac{(1 + y^2)^4}{y^3} S(y),$$

where

$$S(y) = 11y^5 - 20y^4 + 19y^3 - 8y^2 - 2y + 2.$$

For $0 \leq y \leq 1/2$,

$$S(y) \geq 2 - 2y - 8y^2 + 9y^3 \geq \frac{1}{8}.$$

Here the first inequality follows from $y^3(11y^2 - 20y + 10) \geq 0$, and the cubic on the right is decreasing on this interval. For $1/2 \leq y \leq 4/5$, write $y = 2/3 + t$. If $t \geq 0$, then $0 \leq t \leq 2/15$, and

$$\begin{aligned} S(y) &= \frac{58}{243} - \frac{14}{81}t + \frac{250}{27}t^2 + \frac{131}{9}t^3 + \frac{50}{3}t^4 + 11t^5 \\ &\geq \frac{58}{243} - \frac{14}{81} \cdot \frac{2}{15} = \frac{262}{1215} > 0. \end{aligned}$$

If $t = -s \leq 0$, where $0 \leq s \leq 1/6$, then

$$\begin{aligned} S(y) &= \frac{58}{243} + \frac{14}{81}s + \frac{250}{27}s^2 - \frac{131}{9}s^3 + \frac{50}{3}s^4 - 11s^5 \\ &\geq \frac{58}{243} + \frac{14}{81}s + \frac{1465}{216}s^2 > 0. \end{aligned}$$

Thus $h' \leq 0$.

Since $v \leq 5/8$, we have $y < 4/5$, and therefore

$$h(y) \geq h(4/5) = \frac{2432980221}{781250000} > \frac{46}{15}.$$

On the other hand, $N \geq 7$ gives

$$\sqrt{2(2+y^2)}(1+y^{N-2}) \leq \sqrt{\frac{21}{4}} \left(1 + \left(\frac{4}{5}\right)^5\right) < \frac{46}{15};$$

after squaring, the final difference is $52766011/351562500 > 0$. This proves (161) and hence (153). $\square$

The next estimate handles all ζ once v is large.

Lemma 9.3 (Large v). *If $v \geq 5/8$ and $N \geq 7$ is odd, then (153) holds.*

Proof. By (137),

$$T_N < N(N+1)(1-\ell). \quad (162)$$

Using (160), $\ell \leq \sqrt{v}$, and $\sigma = 1 + v$, it is enough, after cancelling $1 - \ell$, to prove

$$\frac{N+2}{\sqrt{2(2+v)}}(1+v-\sqrt{v})(1+v)^{N-1} \geq N(N+1). \quad (163)$$

The left side is increasing in v for $v \geq 5/8$: the factor $1+v-\sqrt{v}$ is increasing for $v > 1/4$, and

$$\frac{d}{dv} \log \frac{(1+v)^{N-1}}{\sqrt{2+v}} = \frac{N-1}{1+v} - \frac{1}{2(2+v)} > 0.$$

After division by $N(N+1)$, its ratio under $N \mapsto N+2$ is at least

$$\left(\frac{13}{8}\right)^2 \frac{N(N+1)(N+4)}{(N+2)^2(N+3)} > 1,$$

because, after clearing denominators, the excess is

$$105N^3 + 397N^2 - 348N - 768 > 0 \quad (N \geq 7).$$

It remains to check $N = 7$, $v = 5/8$. The estimates $\sqrt{10} < 19/6$ and $\sqrt{21} < 23/5$ give

$$\frac{9}{\sqrt{21/4}} \left(\frac{13}{8} - \sqrt{\frac{5}{8}}\right) \left(\frac{13}{8}\right)^6 > \frac{75}{23} \left(\frac{13}{8}\right)^6 > 56,$$

the final excess being $24369203/6029312$. Thus (163) holds. $\square$

9.3 Below the cycle scale: the growth lemma

It remains, except for $N = 7$, to handle $\zeta \leq N$. The compensation lemma reduces this to one variable.

Lemma 9.4 (One-variable growth). *Let $N \geq 9$ be odd. If $0 \leq x < 1$ and*

$$1 - x \geq \frac{2x^2}{N}, \quad (164)$$

then

$$J_N(x) := \frac{N+2}{N}(1-x)^2(1+x^3) \left(\frac{N(1+x^2)}{N-x^2} \right)^{3N/4-1} \geq 1. \quad (165)$$

Proof. Put $A = 3N/4 - 1$.

Step 1: $0 \leq x \leq 2/5$. Taking logarithms in (165), and using

$$\begin{aligned} \log(1 + 2/N) &\geq \frac{2}{N} - \frac{2}{N^2}, \\ \log(1 + x^2) &\geq x^2 - \frac{x^4}{2}, \\ -\log(1 - x^2/N) &\geq \frac{x^2}{N}, \end{aligned}$$

together with

$$2 \log(1 - x) + \log(1 + x^3) \geq -2x - x^2 - \frac{x^4}{1-x} - \frac{x^6}{2}, \quad (166)$$

we obtain

$$\begin{aligned} \log J_N(x) \geq \mathcal{L}_N(x) &:= \frac{2}{N} - \frac{2}{N^2} - 2x \\ &+ \left[A \left(1 + \frac{1}{N} \right) - 1 \right] x^2 - \frac{A}{2} x^4 - \frac{x^4}{1-x} - \frac{x^6}{2}. \end{aligned} \quad (167)$$

For completeness, (166) follows by writing

$$2 \log(1 - x) = -2x - x^2 - \frac{2}{3}x^3 - 2 \sum_{j \geq 4} \frac{x^j}{j} \geq -2x - x^2 - \frac{2}{3}x^3 - \frac{x^4}{1-x}$$

and using $\log(1 + x^3) \geq x^3 - x^6/2$.

On $0 \leq x \leq 1/4$,

$$x^4 \leq \frac{x^2}{16}, \quad \frac{x^4}{1-x} \leq \frac{x^2}{12}, \quad \frac{x^6}{2} \leq \frac{x^2}{512}.$$

Thus

$$\mathcal{L}_N(x) \geq c_N - 2x + B_N x^2,$$

where

$$c_N = \frac{2}{N} - \frac{2}{N^2}, \quad B_N = \frac{1116N^2 - 2003N - 1536}{1536N}.$$

For $N \geq 9$, $B_N > 0$, and

$$B_N c_N - 1 = \frac{348N^3 - 3119N^2 + 467N + 1536}{768N^3} > 0. \quad (168)$$

The numerator equals 6792 at $N = 9$; its derivative is 28889 there and is increasing afterwards. Hence the quadratic has positive global minimum, and $\mathcal{L}_N(x) > 0$ on $[0, 1/4]$.

On $1/4 \leq x \leq 2/5$, differentiation in the real parameter N gives

$$\frac{\partial \mathcal{L}_N}{\partial N} \geq \frac{93}{2048} - \frac{2}{81} > 0.$$

It is therefore enough to take $N = 9$. Direct differentiation gives

$$\mathcal{L}_9(1/4) = \frac{11795}{663552}, \quad \mathcal{L}'_9(1/4) = \frac{1295}{3072}, \quad \mathcal{L}''_9(2/5) = \frac{49}{3375}. \quad (169)$$

Moreover,

$$\mathcal{L}'''_9(x) = -\frac{3xR(x)}{(1-x)^4},$$

where

$$R(x) = 20x^6 - 80x^5 + 143x^4 - 174x^3 + 166x^2 - 104x + 31.$$

For $0 \leq x \leq 2/5$,

$$R(x) \geq 31 - 104x + \frac{2282}{25}x^2 \geq \frac{2503}{625} > 0.$$

Indeed, the difference from the first quadratic is

$$\frac{x^2}{25}(500x^4 - 2000x^3 + 3575x^2 - 4350x + 1868).$$

For $x \leq 2/5$, the polynomial in parentheses is at least

$$500x^4 + (3575 - 800)x^2 + (1868 - 1740) > 0,$$

because $-2000x^3 \geq -800x^2$ and $-4350x \geq -1740$. The quadratic $31 - 104x + (2282/25)x^2$ is decreasing on this interval and has value $2503/625$ at $x = 2/5$. Thus $\mathcal{L}'''_9 < 0$. Since $\mathcal{L}''_9(2/5) > 0$, we have $\mathcal{L}''_9 > 0$ throughout the interval; the first two inequalities in (169) then imply $\mathcal{L}_9 > 0$.

Step 2: $2/5 \leq x \leq \beta_N$. Here β_N is the positive root of

$$1 - \beta_N = \frac{2\beta_N^2}{N};$$

condition (164) is exactly $x \leq \beta_N$. Since $N/(N - x^2) > 1$, it is enough to prove

$$K_N(x) := \frac{N+2}{N}(1-x)^2(1+x^3)(1+x^2)^{3N/4-1} \geq 1. \quad (170)$$

The logarithmic derivative of K_N has the opposite sign to

$$P_N(x) = 3Nx^5 - 3Nx^4 + 3Nx^2 - 3Nx + 6x^5 - 2x^4 \quad (171)$$

$$+ 10x^3 - 6x^2 + 4x + 4. \quad (172)$$

At $x = 2/5$,

$$P_N(2/5) = -\frac{2(1197N - 8266)}{3125} < 0,$$

while

$$P''_N(2/5) = \frac{6(17N + 66)}{25} > 0$$

and

$$P'''_N(x) = 12 [N(15x^2 - 6x) + 30x^2 - 4x + 5] > 0 \quad (x \geq 2/5).$$

Thus P_N is convex on the interval and, starting negative, can cross zero at most once. Therefore K_N has no interior minimum: it is enough to check $x = 2/5$ and $x = \beta_N$.

At $x = 2/5$,

$$K_9(2/5) = \frac{1463}{3125} \left(\frac{29}{25} \right)^{23/4}.$$

Taylor's formula with positive third derivative gives

$$\left(1 + \frac{4}{25} \right)^{23/4} \geq 1 + \frac{23}{4} \frac{4}{25} + \frac{1}{2} \frac{23}{4} \frac{19}{4} \left(\frac{4}{25} \right)^2 = \frac{2837}{1250}.$$

Hence

$$K_9(2/5) \geq \frac{4150531}{3906250} > 1.$$

Furthermore,

$$\frac{K_{N+2}(2/5)}{K_N(2/5)} = \frac{N(N+4)}{(N+2)^2} \left(\frac{29}{25} \right)^{3/2} \geq \frac{117}{121} \frac{31}{25} > 1,$$

where Bernoulli's inequality was used in the last step. Thus the first endpoint is settled for every odd $N \geq 9$.

For the second endpoint, use $(1 - \beta_N)^2 = 4\beta_N^4/N^2$. The function $h_N(x) = 1 - x - 2x^2/N$ is strictly decreasing, and $h_N(\beta_N) = 0$. The exact evaluations

$$h_9(21/25) = \frac{2}{625}, \quad h_{11}(6/7) = \frac{5}{539}, \quad h_{13}(7/8) = \frac{3}{416}$$

show that $\beta_9 > 21/25$, $\beta_{11} > 6/7$, and $\beta_N \geq 7/8$ for every $N \geq 13$. When $N = 9$, hence

$$\begin{aligned} K_9(\beta_9) &> \frac{11}{9} \frac{4}{81} \frac{497}{1000} \frac{159}{100} \left(\frac{17}{10} \right)^5 \frac{37}{25} \\ &= \frac{15221984467459}{15187500000000} > 1. \end{aligned}$$

Here we used

$$\left(\frac{21}{25} \right)^4 > \frac{497}{1000}, \quad 1 + \left(\frac{21}{25} \right)^3 > \frac{159}{100}, \quad 1 + \left(\frac{21}{25} \right)^2 > \frac{17}{10},$$

and

$$\left(\frac{17}{10} \right)^{3/4} > \frac{37}{25}.$$

The last inequality follows after raising both sides to the fourth power.

When $N = 11$, $\beta_{11} > 6/7$, and the coarser estimates $\beta_{11}^4 > 1/2$, $1 + \beta_{11}^3 > 3/2$, and $1 + \beta_{11}^2 > 17/10$ give

$$K_{11}(\beta_{11}) > \frac{13}{11} \frac{4}{121} \frac{1}{2} \frac{3}{2} \left(\frac{17}{10} \right)^7 = \frac{16003208247}{13310000000} > 1.$$

Finally, for $N \geq 13$, $\beta_N \geq 7/8$. Dropping the factors $(N+2)/N$ and $1 + \beta_N^3$,

$$K_N(\beta_N) \geq \frac{4(7/8)^4}{N^2} \left(\frac{113}{64} \right)^{3N/4-1}.$$

At $N = 13$, this is larger than

$$\frac{4(7/8)^4}{13^2} \left(\frac{7}{4} \right)^8 = \frac{13841287201}{11341398016} > 1.$$

The ratio of these lower bounds under $N \mapsto N+2$ is at least

$$\left(\frac{13}{15} \right)^2 \left(\frac{7}{4} \right)^{3/2} > 1.$$

This completes all endpoint checks and the proof of (165). □

Lemma 9.5 (Below the cycle scale). *If $\zeta \leq N$ and $N \geq 9$ is odd, then (153) holds.*

Proof. By Theorem 9.1,

$$\begin{aligned} & (N+2)c_\xi(1-\ell)(\sigma-\ell)\sigma^{N-1} \\ & \geq (N+2)(1-\ell)^2(1+\ell^3)\sigma^{N-1-\zeta/4}. \end{aligned} \quad (173)$$

Indeed, $(1+\ell)(\sigma-\ell) \geq 1+\ell^3$ because $\sigma \geq 1+\ell^2$.

Since $\zeta = \ell^2/u \leq N$,

$$u \geq \frac{\ell^2}{N}, \quad a \leq \frac{N-\ell^2}{N}.$$

Thus

$$\sigma = \frac{1+\ell^2}{a} \geq \frac{N(1+\ell^2)}{N-\ell^2}. \quad (174)$$

Also $N-1-\zeta/4 \geq 3N/4-1$. Finally, $\ell < 2a-1$ from the proof of (137), so

$$1-\ell > 2u = \frac{2\ell^2}{\zeta} \geq \frac{2\ell^2}{N}.$$

Applying Theorem 9.4 with $x = \ell$ to (173) shows that its right side is at least N . Since $T_N < N$, (153) follows. $\square$

9.4 The exceptional cycle scale $N = 7$, i.e. $m = 9$

The preceding lemmas already cover $N = 7$ whenever either $\zeta \geq 7$ or $v \geq 5/8$. It remains to prove (153) under

$$N = 7, \quad \zeta \leq 7, \quad v \leq \frac{5}{8}. \quad (175)$$

Lemma 9.6 (The small- v part of $N = 7$). *Under (175), if $0 < v \leq 1/4$, then (153) holds.*

Proof. Put $s = \sqrt{v}$. Since

$$\frac{\ell^2}{v} = \frac{\zeta}{\zeta+1+v} \leq \frac{7}{8+v} < \frac{7}{8} < \left(\frac{15}{16}\right)^2,$$

we have

$$\ell < \frac{15}{16}s. \quad (176)$$

We next prove

$$\xi \geq \frac{1}{4v}. \quad (177)$$

Let $t = \zeta + v$. Formula (156) becomes

$$4v\xi = \frac{16(t+1)^2}{t^2(t(v+2)+2)}.$$

The reciprocal factor

$$G(t, v) = \frac{t^2(t(v+2)+2)}{(t+1)^2}$$

is increasing in both variables, since

$$\frac{\partial G}{\partial t} = \frac{t(t^2v + 2t^2 + 3tv + 6t + 4)}{(t+1)^3} > 0, \quad \frac{\partial G}{\partial v} = \frac{t^3}{(t+1)^2} > 0.$$

Since $t \leq 7 + 1/4 = 29/4$, its maximum is

$$\frac{246413}{17424} < 16.$$

This proves (177).

The function $(1 + 4\xi)/(2 + 4\xi)$ is increasing in ξ , so (177) gives

$$\varphi \geq \frac{1 + v}{1 + 2v}.$$

Also $\alpha > 1/(2 + v)$, hence

$$c_\xi \sigma^6 \geq \frac{(1 + v)^7}{(1 + 2v)\sqrt{1 + v/2}}.$$

We shall use the elementary lower bound

$$\frac{(1 + v)^7}{(1 + 2v)\sqrt{1 + v/2}} \geq \left(1 - \frac{7v}{8}\right) (1 + 5v + 11v^2). \quad (178)$$

Both sides are positive. If $L(v)$ and $R(v)$ denote respectively the left and right sides of (178), then exact expansion gives

$$\begin{aligned} & 128(1 + 2v)^2(1 + v/2)(L(v)^2 - R(v)^2) \\ &= v(128v^{13} + 1792v^{12} + 11648v^{11} + 46592v^{10} + 128128v^9 \\ &\quad + 232540v^8 + 345884v^7 + 492971v^6 + 464196v^5 \\ &\quad + 258097v^4 + 86924v^3 + 18035v^2 + 2254v + 160), \end{aligned} \quad (179)$$

which is nonnegative.

For the defect, use (122). From (136),

$$\begin{aligned} T_7 &\leq 7 + 7v - (1 - v/2)(1 - v^{7/2}) \\ &\leq 6 + \frac{15}{2}v + \frac{1}{2}v^3, \end{aligned}$$

because $\sqrt{v} \leq 1/2$. On the other hand, (176) and (178) show that the right side of (153) is at least

$$9 \left(1 - \frac{7}{8}s^2\right) \left(1 - \frac{15}{16}s\right) \left(1 + s^2 - \frac{15}{16}s\right) (1 + 5s^2 + 11s^4).$$

After subtracting $6 + 15s^2/2 + s^6/2$, we obtain the polynomial

$$\begin{aligned} P_9(s) &= \frac{10395}{128}s^9 - \frac{333333}{2048}s^8 + \frac{13635}{128}s^7 + \frac{51005}{2048}s^6 \\ &\quad - \frac{18765}{128}s^5 + \frac{264969}{2048}s^4 - \frac{4995}{64}s^3 + \frac{11913}{256}s^2 \\ &\quad - \frac{135}{8}s + 3. \end{aligned} \quad (180)$$

By Theorem A.1, $P_9(s) > 0$ on $[0, 1/2]$. This proves (153). $\square$

Lemma 9.7 (The middle- v part of $N = 7$). *Under (175), if $1/4 \leq v \leq 5/8$, then (153) holds.*

Proof. We first show $\xi > 1/3$. With $t = \zeta + v$,

$$\xi = \frac{4}{v} \frac{(t + 1)^2}{t^2(t(v + 2) + 2)}.$$

The quantity $vG(t, v)$, with G as in the proof of [Theorem 9.6](#), is increasing in both variables. Indeed,

$$\frac{\partial(vG)}{\partial t} = v \frac{\partial G}{\partial t} > 0, \quad \frac{\partial(vG)}{\partial v} = \frac{2t^2(tv + t + 1)}{(t + 1)^2} > 0.$$

At $(t, v) = (61/8, 5/8)$, it equals

$$\frac{26214445}{2437632} < 12.$$

Thus $\xi > 1/3$, and hence

$$\varphi \geq \frac{7}{10}. \quad (181)$$

Put

$$y^2 = \frac{7v}{8 + v}. \quad (182)$$

Since $\zeta \leq 7$, we have $\ell \leq y$. The first half of the compensation proof, [\(155\)](#), gives

$$\sqrt{2\alpha} \geq 1 - \ell^2 \geq 1 - y^2.$$

Also, $v^{7/2} \leq v/2$ on $[1/4, 5/8]$, because $v^{5/2} \leq (5/8)^{5/2} < 1/2$. Together with [\(122\)](#), [\(136\)](#) yields

$$T_7 \leq 6 + 8v. \quad (183)$$

By [\(181\)](#), it is enough to prove

$$\frac{63}{10}(1 - y^2)(1 - y)(1 + v - y)(1 + v)^6 \geq 6 + 8v. \quad (184)$$

We verify that the ratio of the two sides is increasing. Solving [\(182\)](#) gives

$$v = v(y) = \frac{8y^2}{7 - y^2}.$$

For $1/4 \leq v \leq 5/8$,

$$\frac{9}{20} < y < \frac{5}{7}.$$

Define

$$\mathcal{R}(y) = \frac{(1 - y^2)(1 - y)(1 + v(y) - y)(1 + v(y))^6}{6 + 8v(y)}.$$

A direct logarithmic differentiation gives

$$\frac{\mathcal{R}'(y)}{\mathcal{R}(y)} = \frac{2Q_{10}(y)}{(y - 1)(y + 1)(y^2 - 7)(y^2 + 1)(29y^2 + 21)(y^3 + 7y^2 - 7y + 7)}, \quad (185)$$

where

$$Q_{10}(y) = 58y^{10} + 319y^9 - 1793y^8 - 11280y^7 + 9788y^6 \quad (186)$$

$$- 11354y^5 + 350y^4 + 5880y^3 - 6174y^2 + 5635y - 1029. \quad (187)$$

The denominator in [\(185\)](#) is positive on $[9/20, 5/7]$, and [Theorem A.2](#) gives $Q_{10}(y) > 0$ there. Thus $\mathcal{R}$ is increasing.

At $v = 1/4$, one has $y_0 = \sqrt{7/33} < 6/13$. Since the factors involving y decrease with y ,

$$\begin{aligned} & \frac{63}{10}(1 - y_0^2)(1 - y_0)(5/4 - y_0)(5/4)^6 \\ & > \frac{63}{10} \frac{26}{33} \frac{7}{13} \frac{41}{52} \left(\frac{5}{4}\right)^6 = \frac{18834375}{2342912} > 8. \end{aligned}$$

This proves [\(184\)](#) at the left endpoint. The monotonicity of $\mathcal{R}$ proves it throughout the interval. $\square$

Proposition 9.8 (Linear branch). *For every odd $N \geq 7$, assumption (149) implies*

$$R_m \leq C_m \psi(\xi, \rho).$$

Proof. For $N \geq 9$, if $\zeta \leq N$, use Theorem 9.5. If $\zeta \geq N$, use Theorem 9.2 when $v \leq 5/8$, and Theorem 9.3 when $v \geq 5/8$.

For $N = 7$, the same two broad lemmas cover the cases $\zeta \geq 7$ or $v \geq 5/8$. The remaining corner (175) is covered by Theorems 9.6 and 9.7. In every case (153) holds. Combining it with (151), (152), and (130) proves the proposition. $\square$

10 Assembly of the complete proof

Proof of Theorem 1.1. Put $q = 1 - p$.

If $p \leq 1/2$, the right side of (1) is nonpositive, so the claim follows from $t(C_m, W) \geq 0$. Assume henceforth $p > 1/2$.

The cases $m = 3, 5, 7$ follow at every density from Theorems 2.1, 2.3 and 2.4. Let $m \geq 9$ be odd.

If $p \geq 2/3$, or equivalently $q \leq 1/3$, the result is Theorem 4.10. It remains to consider $1/2 < p < 2/3$, equivalently $1/3 < q < 1/2$. By Theorem 1.2, it is enough to treat a step graphon. If the compression A has no eigenvalue above q , Theorem 5.2 proves the theorem. Otherwise there is a unique frontier eigenvalue α , and Theorems 5.4 and 6.3 reduce the problem to the scalar target (104). If $2\rho\xi \leq 1$, Theorem 8.2 proves that target. If $2\rho\xi > 1$, Theorem 9.8 proves it. The two alternatives are exhaustive. The step-graphon approximation returns the result to arbitrary graphons. $\square$

Remark 10.1 (What is finite and what is uniform). *The dense-region argument is uniform in every odd $m \geq 3$. The intermediate-region argument is uniform in every odd $m \geq 9$, except for a small corner at $m = 9$, where two fixed univariate polynomials are certified positive in Section A. There is no length-by-length computation for $m = 11, 13$, and no adaptive interval subdivision anywhere in the proof.*

Remark 10.2 (Equality and stability). *The manuscript is organized to prove the inequality rather than classify all equality cases. A stability analysis would require tracking near-equality in the cycle-to-path deletion, the gamma moment inequality, the Hilbert–Schmidt budget, the forced-variance inequality, and the active Huber witness. The reductions here identify those loci but do not pursue the classification.*

11 Transferable mechanisms and the orthogonal-polynomial viewpoint

This section is not needed for logical completeness. Its purpose is to extract methods that may be useful in other spectral extremal problems.

11.1 One-step stripping: the Schur complement as a Jacobi recursion

Let K be a finite-dimensional self-adjoint operator, let e be a unit vector, and decompose

$$K = \begin{pmatrix} a & h^* \\ h & B \end{pmatrix} \quad \text{on} \quad \mathbb{R}e \oplus e^\perp.$$

The same block inversion used in (28) gives

$$M_{K,e}(z) := \langle e, (I - zK)^{-1}e \rangle = \frac{1}{1 - az - z^2 \langle h, (I - zB)^{-1}h \rangle}. \quad (188)$$

If $h \neq 0$, let ν be the probability spectral measure of B at $h/\|h\|$. Then

$$M_{K,e}(z)^{-1} = 1 - az - \|h\|^2 z^2 M_\nu(z). \quad (189)$$

This is exactly the first step of the Jacobi continued fraction. Iterating on the cyclic Krylov space generated by e produces a tridiagonal Jacobi matrix and its full J-fraction. In the centered, variance-one normalization, (189) becomes

$$1 - M_\mu(z)^{-1} = z^2 M_\nu(z),$$

the classical once-stripped moment relation emphasized by Anshelevich [1]; coefficient stripping for scalar and matrix Jacobi operators is surveyed by Damanik–Pushnitski–Simon [3].

In our graphon proof, $e = \mathbf{1}$, $a = q$, $h = g$, and $B = A$. Thus the complement degree fluctuation g is the first off-diagonal Jacobi coefficient, while A is the once-stripped tail. The proof does not need to tridiagonalize the tail; it retains its spectral measure μ directly. This is useful in infinite-dimensional problems where the first Jacobi step is rigid but the tail is not.

11.2 First returns, irreducible excursions, and logarithmic cycles

Equation (28) has the first-return form

$$M(z) = \frac{1}{1 - H(z)}, \quad H(z) = qz + z^2 R_+(z).$$

Combinatorially, a closed walk from the distinguished state is a sequence of irreducible returns: either a level step of weight q , or a departure into the orthogonal tail followed by a return, encoded by $z^2 R_+$. This is the same structural decomposition that underlies Flajolet’s continued fraction for weighted Motzkin paths [5] and Lehner’s use of Jacobi matrices, lattice paths, and irreducible paths in cumulant expansions [9].

The logarithms $-\log(1 - Y_\pm) = \sum_{r \geq 1} Y_\pm^r / r$ perform a related but cyclic assembly: r excursions are concatenated with the cyclic weight $1/r$. The two-sided identity then applies a parity projection: for odd coefficients, the stripped-tail terms from A and $-A$ cancel. A general design principle suggested by this proof is:

Separate a distinguished state, encode returns by a Schur complement, and combine a model with a signed or complemented companion so that the uncontrolled tail moments cancel in the desired parity class.

This “strip, assemble, cancel” pattern is broader than graphons.

11.3 Dirichlet diagonalization as probabilistic polarization

The calculation in Theorem 4.3 is an instance of the following general lemma.

Proposition 11.1 (Dirichlet polarization). *Fix $r \geq 1$. Suppose*

$$F(\lambda_1, \dots, \lambda_r) = \sum_{j=0}^d c_j h_j(\lambda_1, \dots, \lambda_r),$$

and define

$$\tilde{F}(\ell) = \sum_{j=0}^d c_j \binom{j+r-1}{r-1} \ell^j.$$

If $\Theta \sim \text{Dirichlet}(1, \dots, 1)$, then

$$F(\lambda_1, \dots, \lambda_r) = \mathbb{E} \tilde{F}(\Theta \cdot \lambda).$$

Consequently, for every interval $I \subset \mathbb{R}$,

$$\min_{\lambda \in I^r} F(\lambda) = \min_{\ell \in I} \tilde{F}(\ell).$$

Proof. This is exactly (36), followed by linearity. The minimum statement follows because $\Theta \cdot \lambda \in I$, and the reverse inequality is obtained on the diagonal. $\square$

This lemma may be viewed as a positive, probabilistic version of polarization: the complete homogeneous basis is precisely the basis for which restriction to the diagonal can be inverted by averaging over a simplex. The law of $\Theta \cdot \lambda$ is a Dirichlet average and, in one dimension, is closely related to B-splines and the Hermite–Genocchi formula for divided differences; see de Boor [2]. The key extremal consequence is unusually strong: positivity on a whole spectral cube is equivalent to positivity on the diagonal whenever the polynomial has this basis structure.

11.4 Beta–gamma algebra and Laplace positivity transfer

The passage from (44) to (48) is an instance of a general positive transform. For suitable f , integers $m > r \geq 1$, and $\ell > 0$,

$$\int_0^\infty \frac{s^{r-1}}{(\ell + s)^m} f(s) ds = \frac{\Gamma(r)}{\Gamma(m)} \int_0^\infty t^{m-r-1} e^{-\ell t} \mathbb{E} f(Y/t) dt, \quad Y \sim \text{Gamma}(r, 1). \quad (190)$$

The proof is just the Laplace representation of $(\ell + s)^{-m}$, followed by $y = ts$. Its value is conceptual: a parameter-dependent rational kernel becomes a positive mixture of expectations under a fixed probability family. Therefore any pointwise-in- t expectation inequality for $f(Y/t)$ transfers automatically to every ℓ .

In the present proof, the repeated variables in the complete homogeneous polynomials first produce a beta distribution; the projective coordinate $s = \ell(1 - x)/x$ turns the beta density into a beta-prime kernel; and the Laplace transform reveals a gamma variable. This beta–gamma route is a useful search heuristic whenever repeated spectral parameters and rational powers occur together.

11.5 The gamma Stein identity and a hidden Laguerre equation

Equation (56) is the standard gamma Stein identity; the operator

$$\mathcal{A}_r f(y) = y f'(y) + (r - y) f(y)$$

characterizes the $\text{Gamma}(r, 1)$ law and is widely used in Stein’s method; see, for example, Gaunt [6]. Here it is used not for distributional approximation but as a ladder relation for shifted moments.

There is an exact orthogonal-polynomial shadow. For an integer $k \geq 0$, put

$$F_k(b) = \mathbb{E}(Y - b)^k.$$

Expansion of the moment, or the Rodrigues formula, gives

$$F_k(b) = (-1)^k k! L_k^{(-r-k)}(-b), \quad (191)$$

where $L_k^{(\alpha)}$ is the generalized Laguerre polynomial. For $k = 2j$, the Laguerre differential equation is exactly (57). The parameter $-r - 2j$ lies outside the classical orthogonality range $\alpha > -1$, so we are using the algebraic differential and ladder structure of Laguerre polynomials, not a positive Laguerre orthogonality measure. Standard background is in Szegő [14].

The function G in (63) is a weighted amplitude whose critical points factor explicitly. Proving that its left maximum dominates its right maximum and then using (66) is analogous in spirit to

Sonin–Sturm arguments for comparing successive extrema of solutions to second-order equations. The important transferable move is: construct a first-order multiplier so that the desired linear combination of a polynomial solution and its derivative becomes an integral of a factored derivative; positivity is then reduced to a global maximum comparison.

This explains the connection with the orthogonal-polynomial techniques suggested by the two cited papers. Anshelevich’s once-stripped moment identity is literally our first Schur step, while Lehner’s Jacobi-matrix and Motzkin-path picture explains the return decomposition of the moment series. The later gamma lemma lands in a different classical family—Laguerre—but uses the same general philosophy: a recurrence or differential equation is more effective than coefficientwise estimation of high moments.

11.6 The one-dangerous-mode principle and Huber duality

The intermediate region exhibits a complementary mechanism. The square budget says that two eigenvalues above q would cost more than pq when $q > 1/3$, so there is at most one dangerous mode. Entrywise constraints on the corresponding eigenfunction force a choice: either g couples substantially to that mode, or a quantitatively controlled part of g must lie in the safe subspace. Retaining both alternatives leads to

$$\rho v^2 + (\xi - v + v^2)_+,$$

and minimization over the shape variable v produces the Huber envelope $\psi(\xi, \rho)$. Its dual variable $\lambda \in [0, 1]$ is an active-set multiplier. The witnesses $\lambda = 2\rho\xi$ and $\lambda = 1$ are the quadratic and saturated linear regimes.

This suggests a general recipe for constrained spectral inequalities:

1. use a trace or energy budget to isolate a finite set of dangerous modes;
2. derive geometric constraints that force compensation either in direct coupling or in the safe complement;
3. keep the channels simultaneously rather than estimating them separately;
4. eliminate the shape by convex duality, and choose dual witnesses adapted to small and large activation.

The method should extend to several dangerous modes, where the scalar Huber envelope is replaced by a low-dimensional convex program.

11.7 Why $q = 1/3$ is the natural interface

The same threshold appears for independent structural reasons.

On the dense side, centering at $1/2$ gives $\delta = 1/2 - q$. The gamma moment lemma pays for the odd term in (53) precisely when

$$2\delta \geq \frac{1}{3}, \quad \text{i.e.} \quad q \leq \frac{1}{3}.$$

On the intermediate side, two eigenvalues larger than q would consume more than $2q^2$ of the square budget, and uniqueness is guaranteed precisely when

$$2q^2 > pq = q(1 - q), \quad \text{i.e.} \quad q > \frac{1}{3}.$$

Thus the two mechanisms meet without overlap or gap. This matching strongly suggests that $q = 1/3$ is an intrinsic phase boundary of the proof problem, not an artifact of constants chosen later in the argument.

A The two Bernstein certificates

For completeness, this appendix records the only two fixed polynomial positivity checks used in the proof. The Bernstein-basis principle used here is reviewed in Farouki [4]. We include the coefficients themselves, not merely a reference to a computer calculation.

For an interval $[a, b]$, put

$$t = \frac{x-a}{b-a}, \quad B_{i,n}^{[a,b]}(x) = \binom{n}{i} t^i (1-t)^{n-i} \quad (0 \leq i \leq n).$$

The functions $B_{i,n}^{[a,b]}$ are nonnegative on $[a, b]$ and sum to one. Hence an expansion

$$P(x) = \sum_{i=0}^n \beta_i B_{i,n}^{[a,b]}(x)$$

with all $\beta_i > 0$ proves $P(x) > 0$ throughout the interval. The coefficients can be checked directly as follows. If

$$P(a + (b-a)t) = \sum_{j=0}^n c_j t^j,$$

then

$$\beta_i = \sum_{j=0}^i c_j \frac{\binom{i}{j}}{\binom{n}{j}} \quad (0 \leq i \leq n). \quad (192)$$

Indeed,

$$t^j = \frac{1}{\binom{n}{j}} \sum_{i=j}^n \binom{i}{j} B_{i,n}(t),$$

and substitution gives (192).

Lemma A.1 (The degree-nine certificate). *The polynomial in (180), namely*

$$P_9(s) = \frac{10395}{128} s^9 - \frac{333333}{2048} s^8 + \frac{13635}{128} s^7 + \frac{51005}{2048} s^6 \\ - \frac{18765}{128} s^5 + \frac{264969}{2048} s^4 - \frac{4995}{64} s^3 + \frac{11913}{256} s^2 - \frac{135}{8} s + 3,$$

is positive on $[0, 1/2]$.

Proof. Its degree-nine Bernstein coefficients on $[0, 1/2]$, in increasing order of the index, are

3,	$\frac{33}{16}$,	$\frac{17795}{12288}$,	$\frac{29843}{28672}$,	$\frac{361761}{458752}$,
$\frac{918263}{1376256}$,	$\frac{7142309}{11010048}$,	$\frac{1096445}{1572864}$,	$\frac{390495}{524288}$,	$\frac{361511}{524288}$.

Every numerator and denominator is positive. The Bernstein partition-of-unity argument therefore proves the claim. Formula (192) gives a short exact-arithmetic verification of the displayed list. $\square$

Lemma A.2 (The degree-ten certificate). *The polynomial Q_{10} in (187) is positive on $[9/20, 5/7]$.*

Proof. Its degree-ten Bernstein coefficients on $[9/20, 5/7]$, in increasing order of the index, are

$$\begin{array}{ccc} \frac{3243737479716939}{5120000000000}, & \frac{24439395309190297}{35840000000000}, & \frac{2256629681468091}{31360000000000}, \\ \frac{653354162075429}{8780800000000}, & \frac{3033296182962901}{40336800000000}, & \frac{625950152734243}{8470728000000}, \\ \frac{1151767076423}{1647086000}, & \frac{6182915802683}{9882516000}, & \frac{7481469276}{14706125}, \\ \frac{67938971678}{201768035}, & \frac{26748047904}{282475249}, & \end{array}$$

Again every coefficient is positive. Since the Bernstein basis is nonnegative and sums to one, $Q_{10} > 0$ on the stated interval. The conversion formula (192) verifies the list using only rational arithmetic. $\square$

Remark A.3 (Why this is a proof certificate). *A Bernstein-coefficient list is not a numerical sample. Once the exact identity with the Bernstein expansion is checked, positivity at every point of the whole interval follows from the nonnegativity of the basis functions. In the present proof no subdivision is used: one interval suffices for each of the two polynomials.*

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